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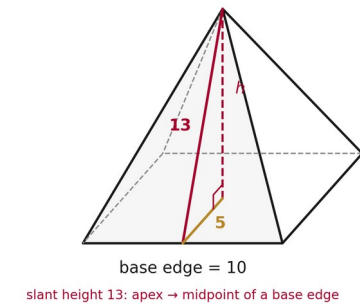
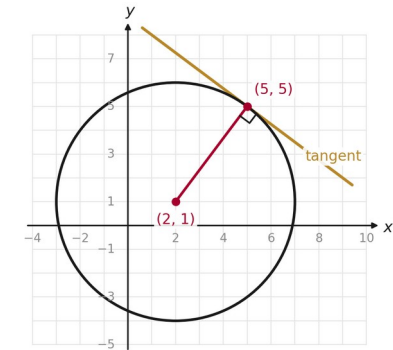
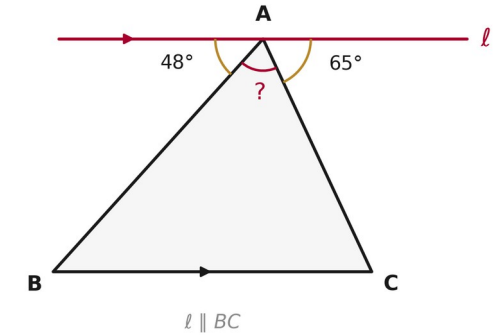
DIGITAL SAT — MATH

Geometry & Trigonometry

Theory Companion — Condensed Rules & Worked Problems

Area and Volume • Lines, Angles & Triangles
Right Triangles & Trigonometry • Circles

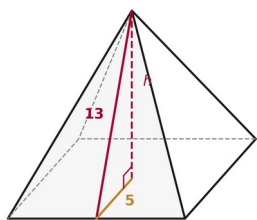
Prepared by the CLC Digital SAT Team



slant height 13: apex $\rightarrow$ midpoint of a base edge

Four Skills, One Domain

Geometry & Trigonometry is one of the four Digital SAT Math domains. Each skill below is condensed to its essential rules, then drilled with 8 Hard / Very Hard worked problems.

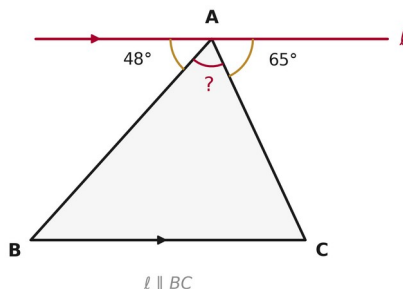


base edge = 10
slant height 13: apex $\rightarrow$ midpoint of a base edge

01

Area and Volume

Composite figures, surface area vs. volume, and how similar solids scale.

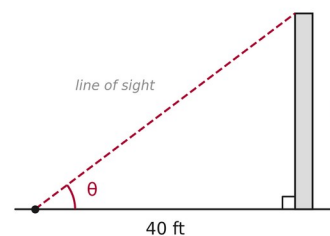


$l \parallel BC$

02

Lines, Angles, and Triangles

Parallel lines, triangle angle facts, similarity, and side-length limits.

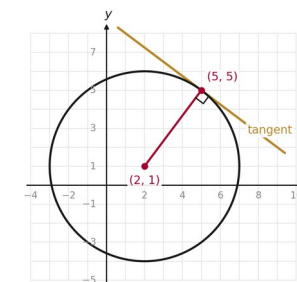


$\tan \theta = 3/4$

03

Right Triangles and Trigonometry

Pythagoras, SOH-CAH-TOA, complementary angles, and special right triangles.



04

Circles

Circle equations, arcs, sectors, radians, inscribed angles, and tangents.

EVERY WORKED PROBLEM IS BROKEN INTO



Formula



Solution



Tip



Trap



Elimination



Shortcut



Desmos

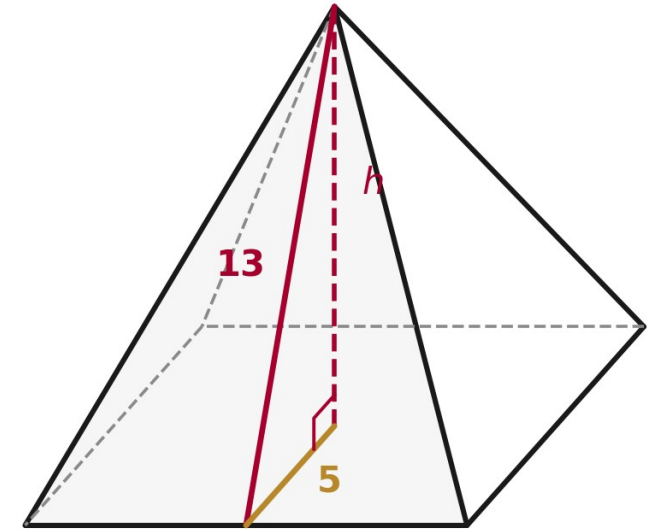
01

Area and Volume

Composite figures, surface area vs. volume, and how similar solids scale.

In this section

- Key terminology — English / Ελληνικά
- Condensed theory with diagrams
- 8 worked problems (Hard / Very Hard)



base edge = 10

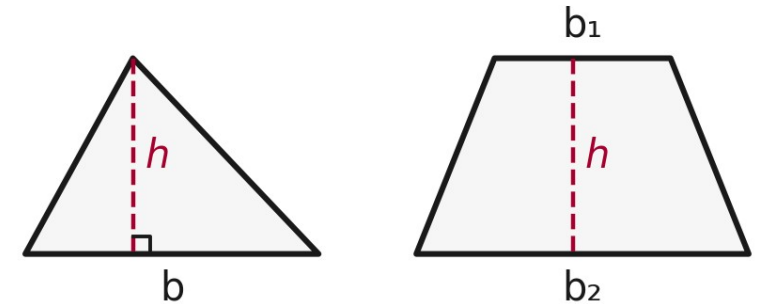
slant height 13: apex $\rightarrow$ midpoint of a base edge

Key Terminology

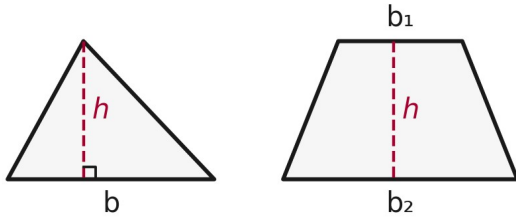
Term (English)	Ελληνικά
Area	Εμβαδόν
Volume	Όγκος
Perimeter	Περίμετρος
Circumference	Περιφέρεια
Surface Area	Επιφάνεια
Base / Height	Βάση / Ύψος
Radius / Diameter	Ακτίνα / Διάμετρος
Similar solids	Όμοια στερεά
Scale factor	Συντελεστής κλίμακας
Composite figure	Σύνθετο σχήμα

A **ἄ** Why it matters

- The formula sheet is given on test day — the skill is choosing the right formula.
- Split composite figures into pieces you have a formula for.
- Know the one rule the sheet never states: how area and volume scale.



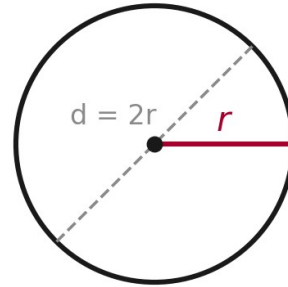
Condensed Theory (1/2)



Triangle & Trapezoid

$$A = \frac{1}{2}bh \quad A = \frac{1}{2}(b_1 + b_2)h$$

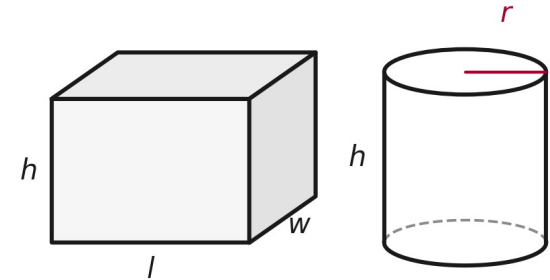
- Height is always perpendicular to the base.
- Trapezoid: average the parallel sides, then $\times$ h.



Circle

$$A = \pi r^2 \quad C = 2\pi r$$

- Check: radius or diameter?
- A dropped factor of 2 is the #1 slip.

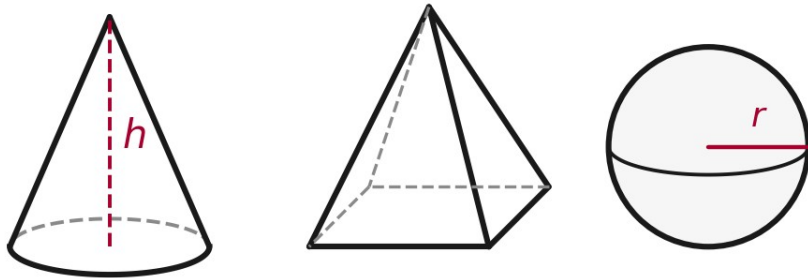


Prism & Cylinder

$$V = lwh \quad V = \pi r^2 h$$

- Both are (base area) $\times$ (height).
- One idea, two formulas.

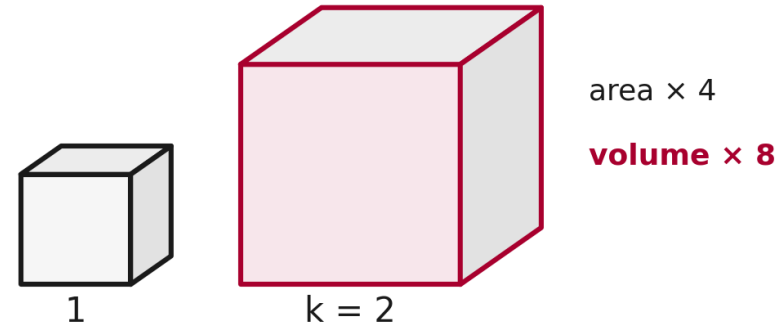
Condensed Theory (2/2)



Cone, Pyramid & Sphere

$$V = \frac{1}{3}\pi r^2 h \quad V = \frac{1}{3}Bh \quad V = \frac{4}{3}\pi r^3$$

- Cone / pyramid = $\frac{1}{3}$ of the matching cylinder / prism.
- Sphere: $SA = 4\pi r^2$.



Similar Solids

$$\text{length} \times k \quad \text{area} \times k^2 \quad \text{volume} \times k^3$$

- Scale every length by k ...
- ... area by k^2 , volume by k^3 — never by k .

Watch Out



Scale factor is NOT on the reference sheet

- Two figures called “similar”? Expect a question built on length^2 vs. length^3 scaling.



Composite figures

- Separate the shape into known pieces, solve each, then add or subtract.
- Never guess a single formula for a combined shape.

Problem 1

Hard

A window is shaped like a rectangle with a semicircle on top, as shown. The rectangle is 30 inches wide and 40 inches tall, and the semicircle's diameter equals the width of the rectangle. What is the total area of the window, in square inches, in terms of π ?

Formula / Rule

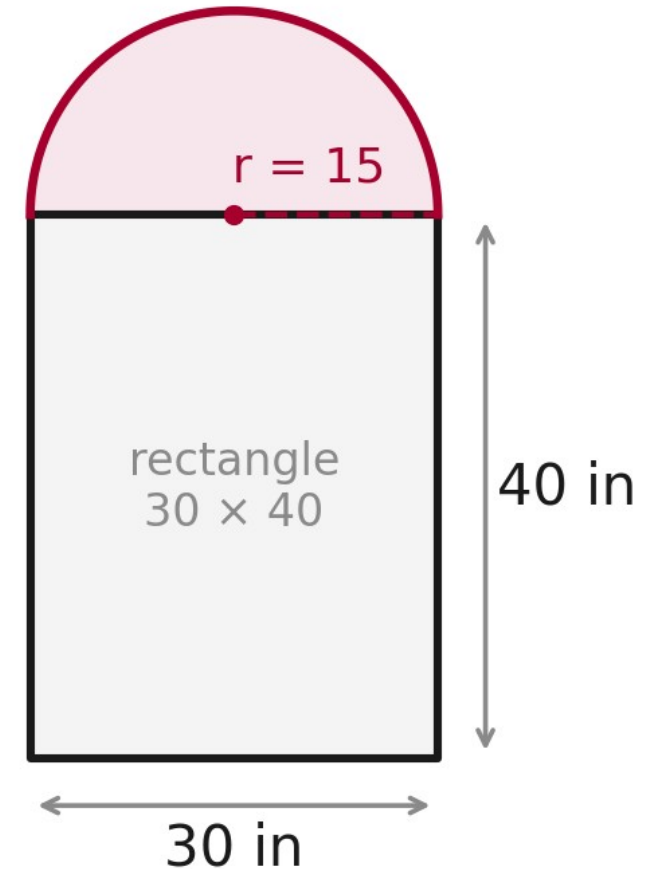
Area of a rectangle; area of a circle (half of it, for the semicircle).

Solution

1. Find the rectangle's area: $A = 30 \times 40 = 1,200$ square inches.
2. The semicircle's diameter is 30, so its radius is 15.
3. Find the full circle's area at that radius: $A = \pi(15)^2 = 225\pi$.
4. The window only has half of that circle, so the semicircle contributes $(1/2)(225\pi) = 112.5\pi$ square inches.
5. Add the two pieces: total area = $1,200 + 112.5\pi$.

✓ **Answer:** $1,200 + 112.5\pi$ square inches

semicircle (diameter 30)





Problem 1 — Strategy & Elimination



Tip

For a composite figure, immediately identify the shared edge (here, the 30-inch width doubling as the semicircle's diameter) — it's usually the bridge that connects the two pieces' formulas.



Trap

A common error is using the full width (30) as the semicircle's radius instead of halving it first, which roughly quadruples the semicircle's contribution to the total.



Shortcut

Since a semicircle's area is always $(1/2)\pi r^2$, you can write it directly as $\pi d^2/8$ in terms of the diameter — here, $\pi(30)^2/8 = 900\pi/8 = 112.5\pi$ — skipping the halving step.



Desmos Tip

There's nothing to graph here, but you can type $1200 + 112.5 \cdot \pi$ directly into Desmos to get the decimal value (≈ 1553.4) as a sanity check on your exact-form answer.



Process of Elimination

Correct answer: **$1,200 + 112.5\pi$ square inches**

Tempting wrong answer	Exactly what went wrong
$1,200 + 450\pi$	Used 30 as the radius instead of the diameter: $\frac{1}{2} \cdot \pi(30)^2 = 450\pi$.
$1,200 + 225\pi$	Found the full circle's area (225π) but forgot to halve it.
$1,312.5\pi$	Treated 1,200 as a multiple of π : $(1,200 + 112.5)\pi$.

Problem 2

Hard

A cylindrical water tank has a radius of 5 feet. The tank's total surface area, including the top and bottom circles, is 220π square feet. What is the volume of the tank, in cubic feet, in terms of π ?

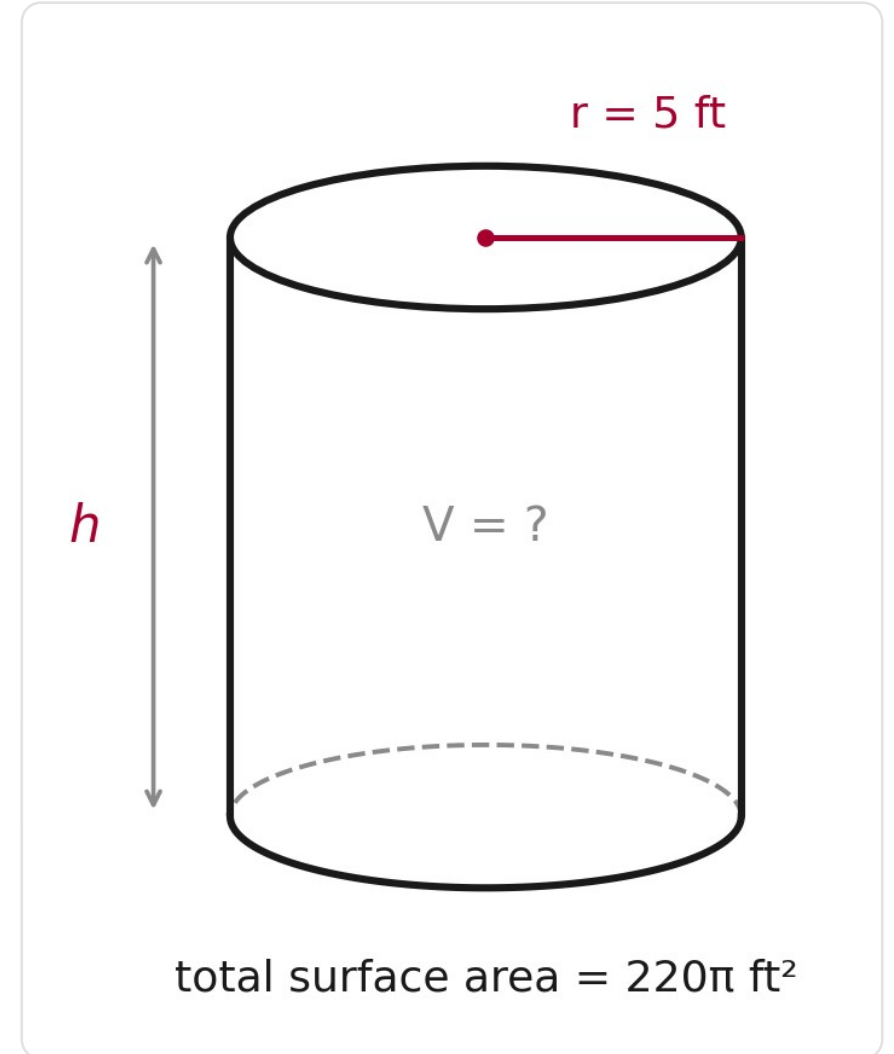
Formula / Rule

Surface area of a cylinder: $SA = 2\pi r^2 + 2\pi rh$. Volume of a cylinder: $V = \pi r^2 h$.

Solution

1. Write the surface area formula with $r = 5$: $SA = 2\pi(5)^2 + 2\pi(5)h = 50\pi + 10\pi h$.
2. Set this equal to the given surface area: $50\pi + 10\pi h = 220\pi$.
3. Subtract 50π from both sides: $10\pi h = 170\pi$.
4. Divide both sides by 10π : $h = 17$.
5. Now find the volume: $V = \pi(5)^2(17) = \pi(25)(17) = 425\pi$.

✓ **Answer:** 425π cubic feet





Problem 2 — Strategy & Elimination



Tip

When a problem gives surface area but asks for volume, treat it as a two-stage problem: solve for the missing dimension first using the surface area equation, then plug that dimension into the volume formula.



Trap

A frequent error is forgetting that a cylinder's surface area includes BOTH circular ends ($2\pi r^2$, not πr^2), which throws off the entire equation used to solve for h .



Shortcut

Once you see the surface-area equation is linear in h , you can skip writing ' $2\pi(5)^2$ ' as a separate step — just compute $2\pi(5)^2 = 50\pi$ mentally as 'twice 25π ' and move straight to isolating h .



Desmos Tip

Type $425 \cdot \pi$ into Desmos to get ≈ 1335.2 cubic feet, and separately confirm $2 \cdot \pi \cdot 25 + 2 \cdot \pi \cdot 5 \cdot 17$ evaluates to the same value as $220 \cdot \pi$.



Process of Elimination

Correct answer: **425π cubic feet**

Tempting wrong answer	Exactly what went wrong
487.5π	Counted only ONE circular end: $25\pi + 10\pi h = 220\pi \rightarrow h = 19.5$.
850π	Found $h = 17$, then used $2\pi r^2 h$ — carried the '2' over from the surface-area formula.
375π	Subtraction slip: $220\pi - 50\pi = 150\pi \rightarrow h = 15$.

Problem 3

Very Hard

Two similar cones have volumes in the ratio $27 : 8$. If the radius of the larger cone is 9 centimeters, what is the radius of the smaller cone?

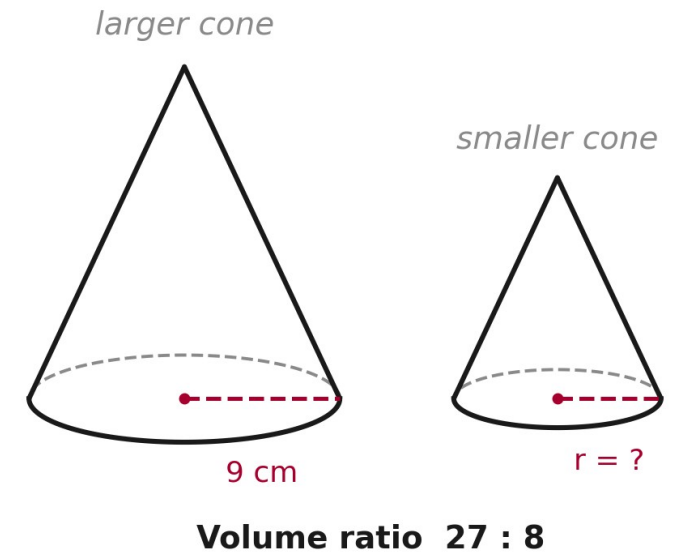
Formula / Rule

For similar solids, volume scale factor = (linear scale factor)³.

Solution

1. Let k be the linear scale factor from the smaller cone to the larger cone.
2. Volumes scale by k^3 , so $k^3 = 27/8$.
3. Take the cube root of both sides: $k = 3/2$.
4. This means every length in the larger cone is $3/2$ times the corresponding length in the smaller cone.
5. So the larger cone's radius equals $(3/2)$ times the smaller cone's radius: $9 = (3/2) \times r$.
6. Solve for r : $r = 9 \div (3/2) = 9 \times (2/3) = 6$.

✓ **Answer:** 6 centimeters





Problem 3 — Strategy & Elimination



Tip

Whenever a ratio of volumes (or areas) is given for similar figures, take the cube root (or square root) immediately to convert it into a linear scale factor — don't try to work with the volume ratio directly.



Trap

A very common error is applying the volume ratio 27:8 directly to the radii, assuming radius scales the same way volume does, which skips the required cube root.



Shortcut

Recognize 27:8 as $3^3:2^3$ by inspection — many SAT similar-solids ratios are set up with small perfect cubes specifically so you can read off the scale factor (3:2) without formal cube-rooting.



Desmos Tip

Confirm by typing $9/(3/2)$ into Desmos to get 6, then check $(9^3)/(6^3)$ simplifies to 27/8 as expected.



Process of Elimination

Correct answer: **6 centimeters**

Tempting wrong answer	Exactly what went wrong
≈ 2.67	Applied the volume ratio straight to the radius: $9 \times 8/27$ — no cube root.
≈ 4.90	Took a square root instead of a cube root: $k = \sqrt{(27/8)} \approx 1.84$.
13.5	Multiplied 9 by $3/2$ instead of dividing — that is an even LARGER cone.

Problem 4

Hard

A cone and a cylinder have equal radii and equal volumes. The height of the cylinder is 4. What is the height of the cone?

Formula / Rule

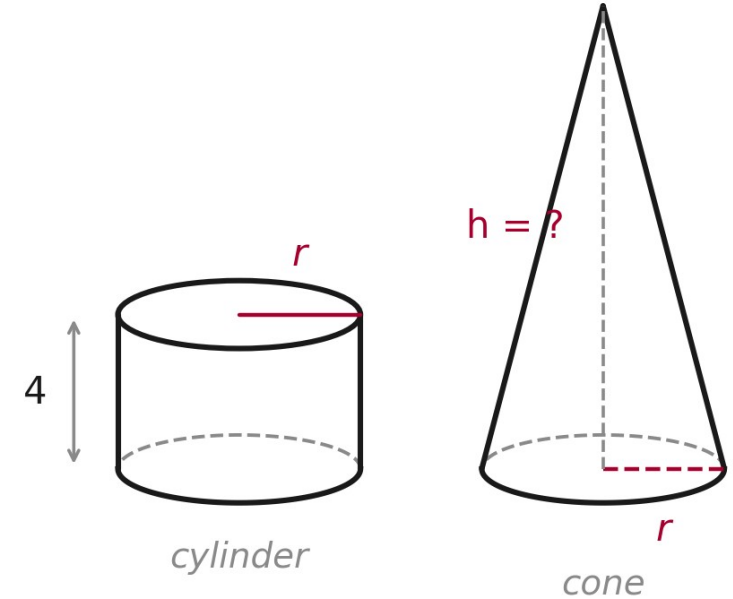
$$V(\text{cylinder}) = \pi r^2 h; \quad V(\text{cone}) = (1/3)\pi r^2 h$$

Solution

1. Let r be the shared radius, h be the cone's height, and set the volumes equal: $(1/3)\pi r^2(h) = \pi r^2(4)$.
2. Divide both sides by πr^2 (it's nonzero, so this is valid and eliminates r entirely): $(1/3)h = 4$.
3. Multiply both sides by 3: $h = 12$.

✓ **Answer: 12**

equal radii • equal volumes





Problem 4 — Strategy & Elimination



Tip

When two solids share both a base measurement and a total volume, set their volume formulas equal to each other and simplify — shared variables like r often cancel out completely, so you don't need their actual value.



Shortcut

Memorize the relationship directly: for equal radius and equal volume, a cone's height is always exactly 3 times a cylinder's height (since the cone formula carries an extra $\frac{1}{3}$). Here, that's immediately $3 \times 4 = 12$.



Trap

Many students assume they're missing information because r is never given numerically, when in fact the problem is specifically built so that r cancels out of the equation.



Desmos Tip

Not needed for this one — it's a pure formula-cancellation problem with no graphable relationship, so the fastest check is simply recomputing $(\frac{1}{3})\pi r^2(12)$ and $\pi r^2(4)$ for any value of r (say $r = 1$) and confirming both equal the same number.



Process of Elimination

Correct answer: **12**

Tempting wrong answer	Exactly what went wrong
4	Assumed equal volume + equal radius means equal height — ignored the cone's $\frac{1}{3}$.
$\frac{4}{3}$ (≈ 1.33)	Computed $\frac{1}{3} \times 4$ instead of solving $\frac{1}{3} \cdot h = 4$.
36	Multiplied by 3 twice.

Problem 5

Very Hard

A sphere has a surface area of 144π square inches. What is the volume of the sphere, in cubic inches, in terms of π ?

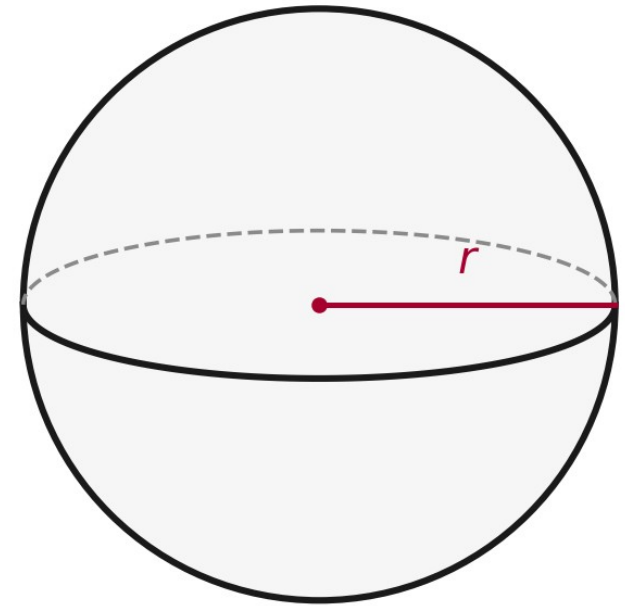
Formula / Rule

Surface area of a sphere: $SA = 4\pi r^2$. Volume of a sphere: $V = (4/3)\pi r^3$.

Solution

1. Set the surface area formula equal to the given value: $4\pi r^2 = 144\pi$.
2. Divide both sides by 4π : $r^2 = 36$.
3. Take the square root: $r = 6$ (radius must be positive).
4. Substitute $r = 6$ into the volume formula: $V = (4/3)\pi(6)^3$.
5. Compute $6^3 = 216$, so $V = (4/3)\pi(216)$.
6. Simplify: $(4/3)(216) = 4 \times 72 = 288$, so $V = 288\pi$.

 **Answer:** 288π cubic inches



surface area = $144\pi \text{ in}^2$



Problem 5 — Strategy & Elimination



Tip

When a sphere problem gives surface area but asks for volume (or vice versa), always solve for the radius first — it's the only bridge between the two formulas.



Shortcut

Once you know $r = 6$, compute $6^3 = 216$ and then multiply by 4 before dividing by 3 ($4 \times 216 = 864$, then $864/3 = 288$) — it's easier mental math than multiplying by the fraction $4/3$ directly.



Trap

A common error is dividing 144π by 4π correctly to get $r^2 = 36$, but then plugging $r^2 = 36$ directly into the volume formula instead of first taking the square root to get $r = 6$.



Desmos Tip

Type $(4/3) * \pi * 6^3$ into Desmos to confirm it returns 288π (≈ 904.8), and separately confirm $4 * \pi * 6^2$ returns 144π to check your radius.



Process of Elimination

Correct answer: **288π cubic inches**

Tempting wrong answer	Exactly what went wrong
48π	Used $r^2 = 36$ in place of r : $(4/3)\pi(36)$.
864π	Multiplied 216 by 4 and forgot to divide by 3.
216π	Dropped the $4/3$ entirely: πr^3 .

Problem 6

Hard

The parallel sides of a trapezoid have lengths of $(2x + 3)$ and $(4x - 1)$, and the height of the trapezoid is 6. If the area of the trapezoid is 96, what is the value of x ?

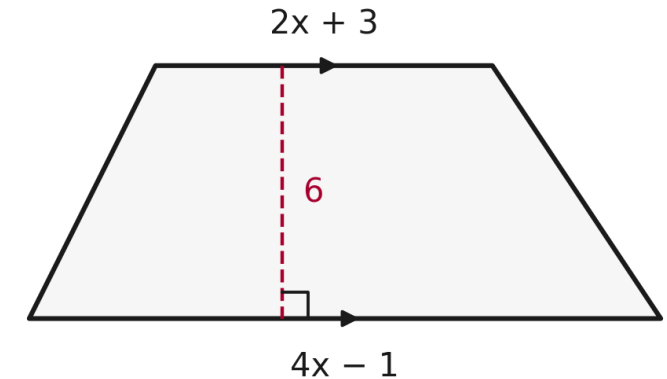
Formula / Rule

Area of a trapezoid: $A = (1/2)(b_1 + b_2)h$.

Solution

1. Substitute the given values into the trapezoid area formula: $96 = (1/2)[(2x + 3) + (4x - 1)](6)$.
2. Combine like terms inside the brackets: $(2x + 3) + (4x - 1) = 6x + 2$.
3. Rewrite the equation: $96 = (1/2)(6x + 2)(6)$.
4. Simplify $(1/2)(6) = 3$, so: $96 = 3(6x + 2)$.
5. Distribute: $96 = 18x + 6$.
6. Subtract 6 from both sides: $90 = 18x$.
7. Divide both sides by 18: $x = 5$.

✓ **Answer:** $x = 5$



Area = 96



Problem 6 — Strategy & Elimination



Tip

When both parallel sides are algebraic expressions, combine them into a single expression first (as if finding their sum), then substitute that single simplified expression into the area formula — it keeps the equation from getting cluttered.



Shortcut

Since $(1/2) \times \text{height}$ simplifies to a clean whole number whenever the height is even, compute that product first (here, $(1/2)(6) = 3$) and multiply it directly by the combined side-length expression, skipping an extra fraction step.



Trap

A common error is multiplying $(1/2)$ and the height (6) separately with only one of the two side-length expressions instead of with their full combined sum, effectively dropping one of the two parallel sides from the equation.



Desmos Tip

Graph $y = (1/2)(6x + 2)(6)$ and $y = 96$ on Desmos and confirm they intersect at $x = 5$.



Process of Elimination

Correct answer: **x = 5**

Tempting wrong answer	Exactly what went wrong
$x = 7/3$	Forgot the $\frac{1}{2}$: $6(6x + 2) = 96$.
$x = 8.25$	Used only one base: $3(4x - 1) = 96$.
$x \approx 31.67$	Forgot the height: $\frac{1}{2}(6x + 2) = 96$.

Problem 7

Very Hard

A square pyramid has a base with side length 10 and a slant height of 13, measured along a face from the midpoint of a base edge to the apex. What is the volume of the pyramid?

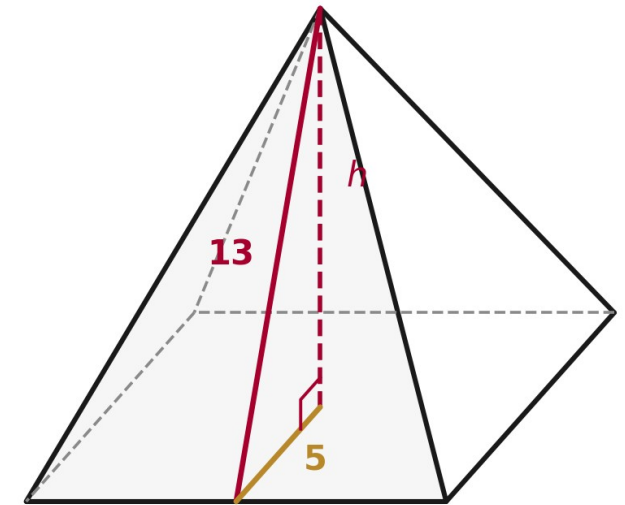
Formula / Rule

Volume of a pyramid: $V = (1/3)(\text{base area})(\text{height})$. The pyramid's actual height, slant height, and half the base edge form a right triangle.

Solution

1. The slant height (13), half the base edge ($10 \div 2 = 5$), and the pyramid's true vertical height (h) form a right triangle, with the slant height as the hypotenuse.
2. Apply the Pythagorean theorem: $5^2 + h^2 = 13^2$.
3. Compute the squares: $25 + h^2 = 169$.
4. Subtract 25 from both sides: $h^2 = 144$.
5. Take the square root: $h = 12$.
6. Find the base area: since the base is a square with side 10, base area = $10^2 = 100$.
7. Substitute into the volume formula: $V = (1/3)(100)(12)$.
8. Simplify: $(100)(12) = 1,200$, and $1,200 \div 3 = 400$.

✓ **Answer: 400**



base edge = 10

slant height 13: apex → midpoint of a base edge



Problem 7 — Strategy & Elimination



Tip

Whenever a pyramid problem gives you a slant height instead of the true vertical height, look for the hidden right triangle formed by the slant height, the vertical height, and half of the relevant base edge — the Pythagorean theorem bridges the two.



Trap

A frequent error is plugging the slant height (13) straight into the volume formula instead of the pyramid's true vertical height.



Shortcut

Recognize 5-12-13 as a Pythagorean triple the moment you see 'half the base edge is 5' and 'slant height is 13' — the vertical height 12 follows immediately without needing to compute the square root by hand.



Desmos Tip

Type $\text{sqrt}(13^2 - 5^2)$ into Desmos to instantly confirm the pyramid's height is 12, then $(1/3) \cdot 100 \cdot 12$ to confirm the volume is 400.



Process of Elimination

Correct answer: 400

Tempting wrong answer	Exactly what went wrong
≈ 433.3	Used the slant height 13 as the height: $\frac{1}{3}(100)(13)$.
1,200	Found $h = 12$ but forgot the $\frac{1}{3}$: 100×12 .
≈ 277	Used the full edge 10 (not 5) in the Pythagorean step: $h = \sqrt{69} \approx 8.3$.

Problem 8

Hard

A rectangular prism has length 6, width 4, and height h , and a volume of 96. If the length is increased to 9 and the width is increased to 6, while the height stays the same, what is the new volume of the prism?

Formula / Rule

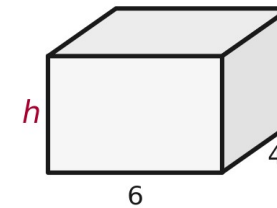
Volume of a rectangular prism: $V = lwh$.

Solution

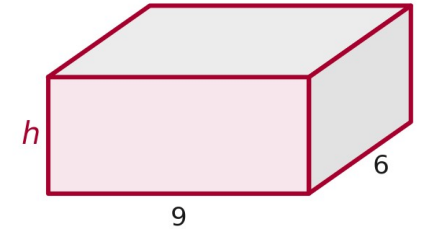
1. Use the original dimensions to solve for h : $6 \times 4 \times h = 96$.
2. Simplify: $24h = 96$.
3. Divide both sides by 24: $h = 4$.
4. Now use the new length and width, with the same height, to find the new volume: $V = 9 \times 6 \times h$.
5. Compute: $9 \times 6 = 54$, and $54 \times 4 = 216$.

✓ **Answer: 216**

BEFORE: $V = 96$



AFTER: $V = ?$



height h stays the same



Problem 8 — Strategy & Elimination



Tip

When a prism problem changes some dimensions but keeps others fixed, first solve for the unknown (fixed) dimension using the original information — you'll need it again for the new volume.



Trap

A common error is assuming the height must also have changed proportionally along with the length and width, when the problem explicitly states the height 'stays the same.'



Shortcut

Since length scaled by a factor of $9/6 = 3/2$ and width scaled by $6/4 = 3/2$, the base area scaled by $(3/2)^2 = 9/4$ — so the new volume is simply $96 \times 9/4 = 216$, without ever solving for h directly.



Desmos Tip

Type $6 \cdot 4 \cdot x = 96$ into Desmos (solving for x) to confirm $h = 4$, then compute $9 \cdot 6 \cdot 4$ to confirm the new volume is 216.



Process of Elimination

Correct answer: **216**

Tempting wrong answer	Exactly what went wrong
144	Updated the length to 9 but kept the width at 4: $9 \times 4 \times 4$.
96	Assumed the volume was unchanged because the height didn't change.
324	Scaled all THREE dimensions by $3/2$: $96 \times (3/2)^3$ — but the height stayed the same.

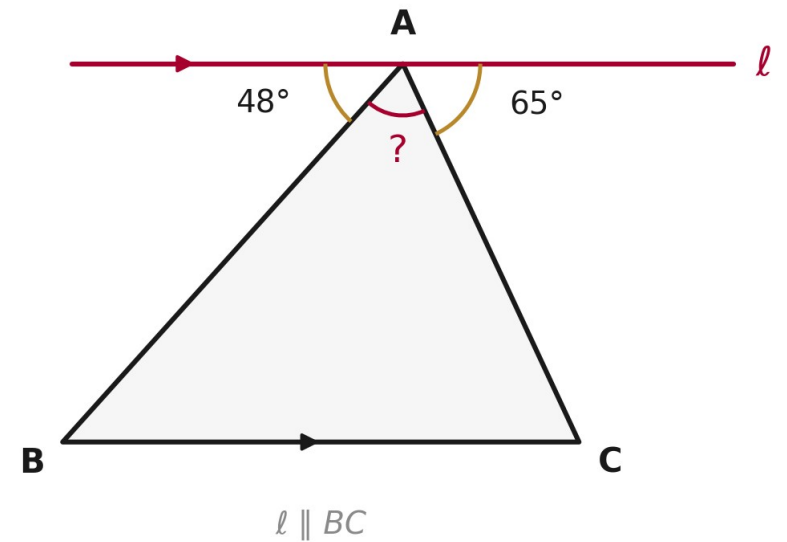
02

Lines, Angles, and Triangles

Parallel lines, triangle angle facts, similarity, and side-length limits.

In this section

- Key terminology — English / Ελληνικά
- Condensed theory with diagrams
- 8 worked problems (Hard / Very Hard)

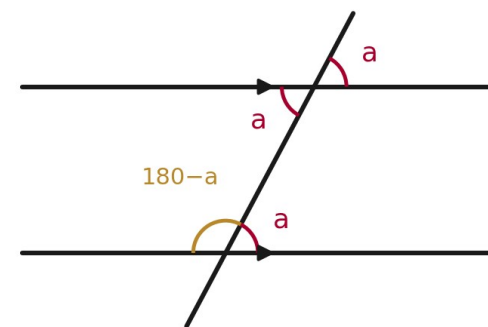


Key Terminology

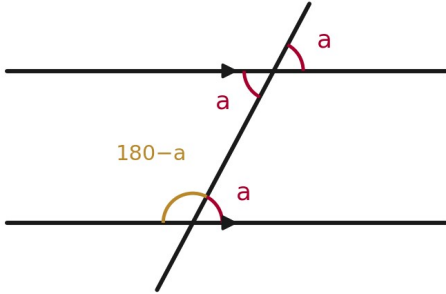
Term (English)	Ελληνικά
Transversal	Διατέμνουσα
Vertical angles	Κατακόρυφες γωνίες
Alternate interior angles	Εναλλάξ εσωτερικές γωνίες
Corresponding angles	Αντίστοιχες γωνίες
Supplementary angles	Παραπληρωματικές γωνίες
Exterior angle	Εξωτερική γωνία
Isosceles triangle	Ισοσκελές τρίγωνο
Similar triangles	Όμοια τρίγωνα
Congruent	Ισομετρικά (ίσα)
Triangle inequality	Τριγωνική ανισότητα

A ✎ Why it matters

- Two rule families: parallel-line angles and triangle properties.
- Hard questions combine two rules in one figure.
- Name the rule for each piece of the figure before writing algebra.



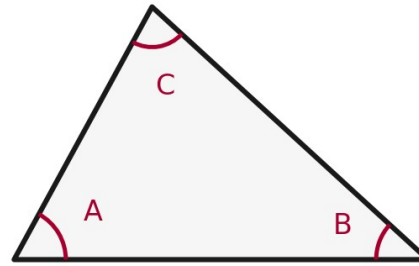
Condensed Theory (1/2)



Parallel Lines + Transversal

corresponding = equal

- Alternate interior angles: equal.
- Same-side interior angles: sum to 180° .

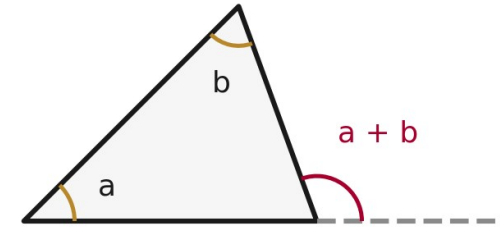


$$A + B + C = 180^\circ$$

Triangle Angle Sum

$$\angle A + \angle B + \angle C = 180^\circ$$

- True for every triangle.
- The most-used fact in this skill.

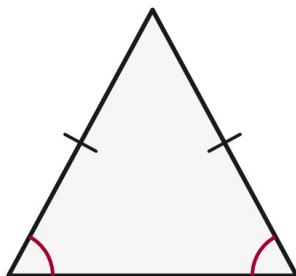


Exterior Angle Theorem

exterior = sum of 2 remote interiors

- Skip finding the third angle.
- Go straight to the equation.

Condensed Theory (2/2)

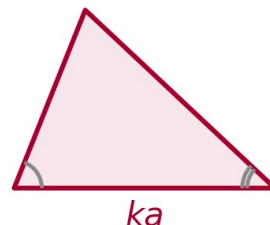
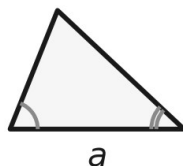


equal sides $\leftrightarrow$ equal base angles

Isosceles Triangle

equal sides $\leftrightarrow$ equal base angles

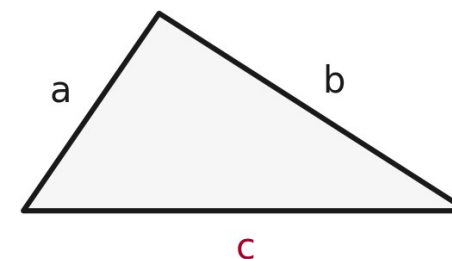
- The equal angles sit OPPOSITE the equal sides.
- Works in both directions.



Similar Triangles (AA)

2 equal angles $\rightarrow$ sides proportional

- Similar means proportional, not equal.
- Match vertices in order: $A \leftrightarrow D$, $B \leftrightarrow E$.



$$|a - b| < c < a + b$$

Triangle Inequality

$$|a - b| < c < a + b$$

- Strict inequalities — endpoints excluded.
- Equal to the sum = a flat line, not a triangle.

Watch Out



“Similar” never means “equal”

- Similar triangles have proportional sides.
- Write the vertex match first: AB/DE , not AB/EF , when $A \leftrightarrow D$ and $B \leftrightarrow E$.



Parallel line through a vertex

- A line through one vertex, parallel to the opposite side, turns alternate interior angles into the triangle's own angles.

Problem 1

Hard

In the figure, lines ℓ and m are parallel and are cut by transversal t . One angle formed at the intersection with ℓ measures $(3x + 10)^\circ$, and its corresponding angle at the intersection with m measures $(5x - 30)^\circ$. What is the measure of the angle at ℓ ?

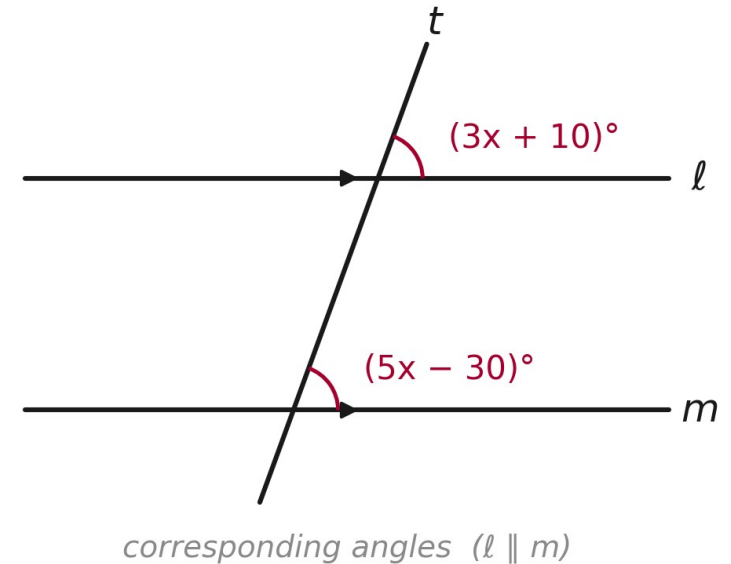
Formula / Rule

Corresponding angles formed by a transversal across parallel lines are equal.

Solution

1. Since $\ell \parallel m$, the two marked angles are corresponding angles, so they're equal: $3x + 10 = 5x - 30$.
2. Subtract $3x$ from both sides: $10 = 2x - 30$.
3. Add 30 to both sides: $40 = 2x$.
4. Divide by 2: $x = 20$.
5. Substitute back into the angle at ℓ : $3(20) + 10 = 60 + 10 = 70$.

✓ **Answer:** 70°





Problem 1 — Strategy & Elimination



Tip

The moment you see two lines marked parallel and a transversal crossing both, scan the figure for which specific angle relationship applies (corresponding, alternate interior, or co-interior) before writing any equation — that choice determines whether the angles are equal or supplementary.



Trap

A common error is treating the two given angles as supplementary (summing to 180°) instead of equal, which happens when a student misidentifies corresponding angles as co-interior angles.



Shortcut

Once you confirm the relationship is 'equal' (not supplementary), you can skip drawing the full figure and just solve the two expressions as a direct linear equation — corresponding/alternate-interior always reduces to 'set equal,' co-interior always reduces to 'set the sum to 180.'



Desmos Tip

There's no graph needed, but you can verify by typing $3 \cdot 20 + 10$ and $5 \cdot 20 - 30$ separately into Desmos to confirm both equal 70.



Process of Elimination

Correct answer: **70°**

Tempting wrong answer	Exactly what went wrong
85°	Treated the angles as supplementary: $8x - 20 = 180 \rightarrow x = 25$.
60°	Found $x = 20$ but dropped the $+10$: $3(20)$.
40°	Stopped at $2x = 40$ and reported that value.

Problem 2

Hard

In triangle ABC, the exterior angle at vertex C measures $(5x - 4)^\circ$. The remote interior angles, at vertices A and B, measure $(2x + 10)^\circ$ and $(x + 14)^\circ$, respectively. What is the measure of the exterior angle at vertex C?

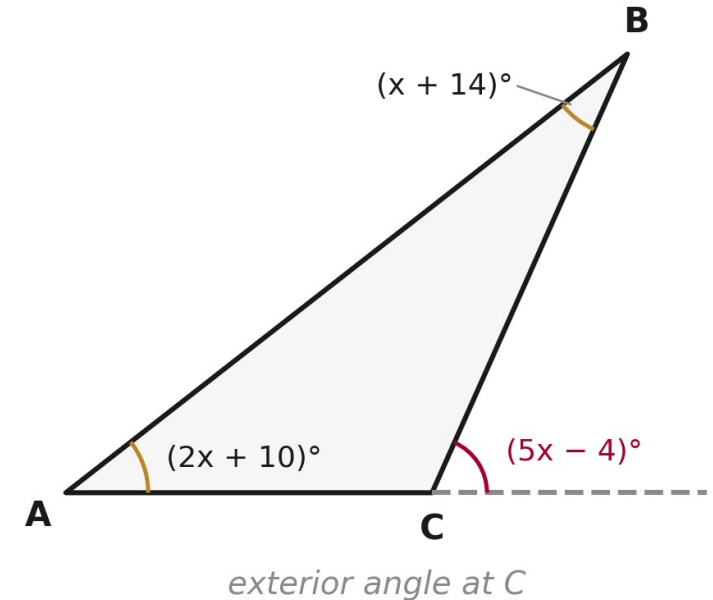
Formula / Rule

Exterior Angle Theorem: an exterior angle equals the sum of the two remote (non-adjacent) interior angles.

Solution

1. Apply the Exterior Angle Theorem: $5x - 4 = (2x + 10) + (x + 14)$.
2. Simplify the right side: $(2x + 10) + (x + 14) = 3x + 24$.
3. Rewrite the equation: $5x - 4 = 3x + 24$.
4. Subtract $3x$ from both sides: $2x - 4 = 24$.
5. Add 4 to both sides: $2x = 28$.
6. Divide by 2: $x = 14$.
7. Substitute into the exterior angle expression: $5(14) - 4 = 70 - 4 = 66$.

 **Answer:** 66°





Problem 2 — Strategy & Elimination



Tip

The Exterior Angle Theorem lets you skip finding the third interior angle and subtracting from 180° — go straight from 'exterior angle = sum of the two remote interior angles' to setting up the equation.



Trap

A frequent error is treating all three expressions as the triangle's interior angles and setting their sum equal to 180° — but $(5x - 4)^\circ$ is an EXTERIOR angle.



Shortcut

You never need to find the third interior angle (at C) explicitly — the exterior angle theorem gives you a direct equation using only the two remote angles, saving a step compared to using the full 180° triangle sum.



Desmos Tip

Confirm by typing $(2 \cdot 14 + 10) + (14 + 14)$ into Desmos — it should equal 66, matching $5 \cdot 14 - 4$.



Process of Elimination

Correct answer: **66°**

Tempting wrong answer	Exactly what went wrong
96°	Treated all three expressions as interior angles: $8x + 20 = 180 \rightarrow x = 20$.
114°	Found the INTERIOR angle at C ($180^\circ - 66^\circ$), not the exterior angle.
38°	Reported angle A instead: $2(14) + 10$.

Problem 3

Hard

Triangle ABC is similar to triangle DEF, with angle A corresponding to angle D and angle B corresponding to angle E. If $AB = 8$, $BC = 12$, $DE = 10$, and $EF = x$, what is the value of x ?

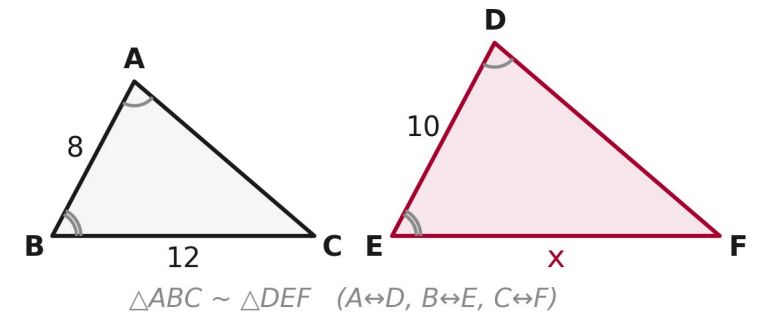
Formula / Rule

For similar triangles, corresponding sides are proportional.

Solution

1. Identify the corresponding sides: AB corresponds to DE (both are between the angle-A/angle-D and angle-B/angle-E vertices), and BC corresponds to EF (both are between the angle-B/angle-E and angle-C/angle-F vertices).
2. Set up the proportion: $AB/DE = BC/EF$.
3. Substitute the known values: $8/10 = 12/x$.
4. Cross-multiply: $8x = 10 \times 12 = 120$.
5. Divide both sides by 8: $x = 15$.

✓ **Answer: 15**





Problem 3 — Strategy & Elimination



Tip

Before setting up any proportion, explicitly write out which vertex corresponds to which (here, $A \leftrightarrow D$, $B \leftrightarrow E$, and therefore $C \leftrightarrow F$) — this makes it much harder to accidentally pair up the wrong sides.



Trap

A common error is pairing AB with EF (or BC with DE) instead of matching sides that connect the SAME pair of corresponding vertices, which produces a proportion with mismatched sides.



Shortcut

Since 8 and 10 don't share an obvious common factor with 12, just cross-multiply directly rather than searching for a scale factor — for messier ratios, cross-multiplication is faster than trying to simplify $8/10$ first.



Desmos Tip

Type $10 \cdot 12 / 8$ directly into Desmos to instantly get 15 without writing out the cross-multiplication by hand.



Process of Elimination

Correct answer: 15

Tempting wrong answer	Exactly what went wrong
9.6	Mismatched the proportion: $8/10 = x/12$.
≈ 6.67	Paired AB with EF: $8/x = 12/10$.
14	Added instead of scaling: DE is 2 more than AB, so $EF = 12 + 2$.

Problem 4

Hard

In isosceles triangle PQR, $PQ = PR$, and the base angles at Q and R each measure $(3x + 5)^\circ$. If the vertex angle at P measures $(4x - 10)^\circ$, what is the measure of angle P?

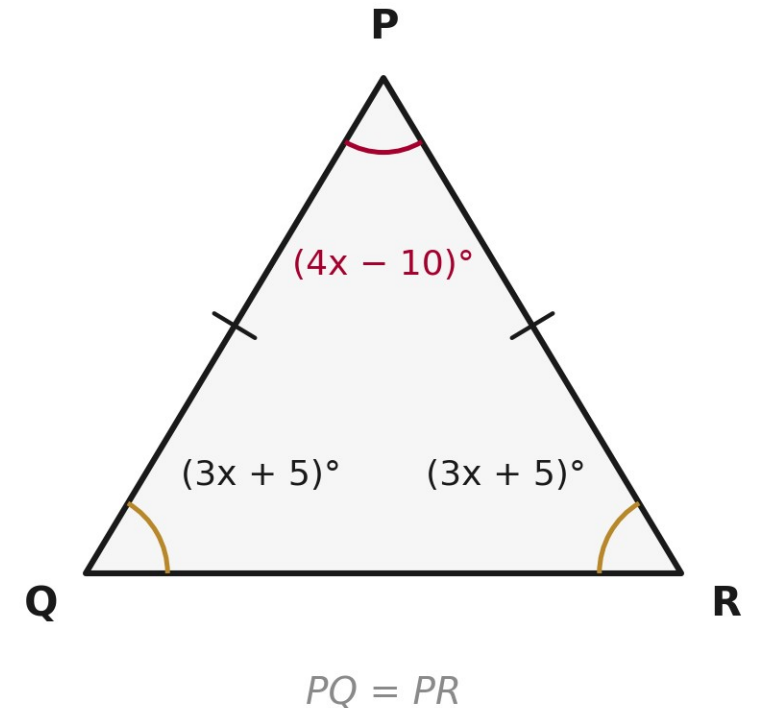
Formula / Rule

Isosceles Triangle Theorem (equal sides give equal base angles) combined with the Triangle Angle Sum Theorem.

Solution

1. Since $PQ = PR$, the triangle is isosceles with the base angles at Q and R equal — this is already given as $(3x+5)^\circ$ for both.
2. Apply the Triangle Angle Sum Theorem: $(3x + 5) + (3x + 5) + (4x - 10) = 180$.
3. Combine like terms: $3x + 3x + 4x + 5 + 5 - 10 = 180$, which simplifies to $10x + 0 = 180$.
4. Divide both sides by 10: $x = 18$.
5. Substitute into the vertex angle expression: $4(18) - 10 = 72 - 10 = 62$.

Answer: 62°





Problem 4 — Strategy & Elimination



Tip

Whenever a problem tells you which two sides of a triangle are equal, immediately mark the two angles **OPPOSITE** those sides as equal too — that's the piece of information the Isosceles Triangle Theorem is really giving you, often before the algebra even starts.



Trap

A common error is assuming the angle AT vertex P (between the two equal sides PQ and PR) is one of the equal base angles, when the equal angles are always the ones **OPPOSITE** the equal sides — i.e., at Q and R, not at P.



Shortcut

Since both base angles use the identical expression $(3x+5)$, you can write the angle sum as $2(3x+5) + (4x-10) = 180$ directly, without spelling out the addition of two identical terms.



Desmos Tip

Graph $y = 2 \cdot (3x+5) + (4x-10)$ and $y = 180$ in Desmos and confirm the intersection is at $x = 18$.



Process of Elimination

Correct answer: **62°**

Tempting wrong answer	Exactly what went wrong
72°	Found $x = 18$ but dropped the -10 : $4(18)$.
59°	Reported a base angle: $3(18) + 5$.
50°	Assumed all three angles equal: $3x + 5 = 4x - 10 \rightarrow x = 15$.

Problem 5

Very Hard

In the figure, line ℓ passes through vertex A of triangle ABC and is parallel to side BC. The angle between line ℓ and segment AB measures 48° , and the angle between line ℓ and segment AC measures 65° , with both angles opening away from the triangle on either side of A. What is the measure of angle A (angle BAC) in the triangle?

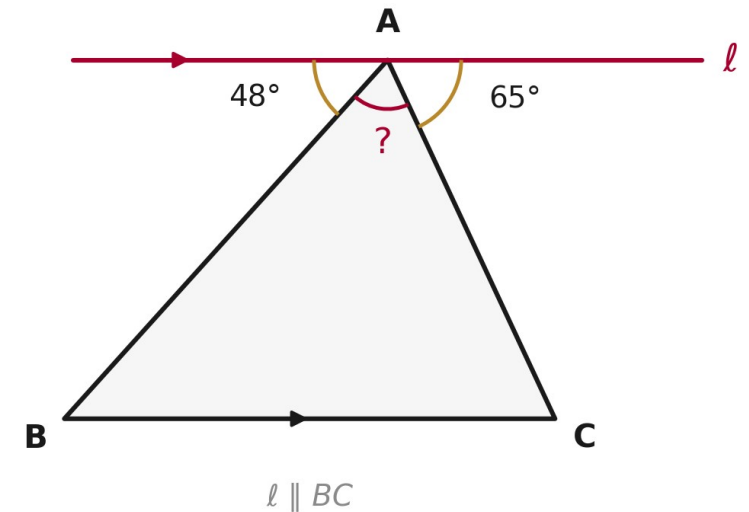
Formula / Rule

Alternate interior angles (from $\ell \parallel BC$) combined with the fact that angles on a straight line sum to 180° .

Solution

1. Since ℓ is a straight line through A, the three angles along it at point A — the 48° angle, angle BAC, and the 65° angle — together make a straight angle, so they sum to 180° .
2. Set up the equation: $48 + (\text{angle BAC}) + 65 = 180$.
3. Combine the known values: $113 + (\text{angle BAC}) = 180$.
4. Subtract 113 from both sides: angle BAC = 67.

✓ **Answer:** 67°





Problem 5 — Strategy & Elimination



Tip

When a figure shows a line drawn through a triangle's vertex parallel to the opposite side, the three angles sitting along that line at the vertex ALWAYS sum to 180° — this is exactly the classical proof that a triangle's angles sum to 180° , dressed up as a Digital SAT question.



Trap

A common error is treating 48° and 65° as angle A's neighbors in the TRIANGLE and stopping after adding them — angle A is the third angle along line ℓ .



Shortcut

Recognize immediately that alternate interior angles make the 48° and 65° angles equal to angles B and C of the triangle respectively — so this problem is secretly just 'angle A = $180 - \text{angle B} - \text{angle C}$,' the ordinary Triangle Angle Sum Theorem, without ever having to prove why.



Desmos Tip

Not a graphing problem, but you can confirm arithmetic by typing $180 - 48 - 65$ into Desmos to instantly get 67.



Process of Elimination

Correct answer: **67°**

Tempting wrong answer	Exactly what went wrong
113°	Added the two given angles ($48^\circ + 65^\circ$) and stopped.
132°	Subtracted only one given angle: $180^\circ - 48^\circ$.
48°	Assumed angle A equals the adjacent 48° angle on line ℓ .

Problem 6

Very Hard

In triangle XYZ, $XY = 11$ and $YZ = 17$. If the length of side XZ is an integer, how many possible values are there for XZ?

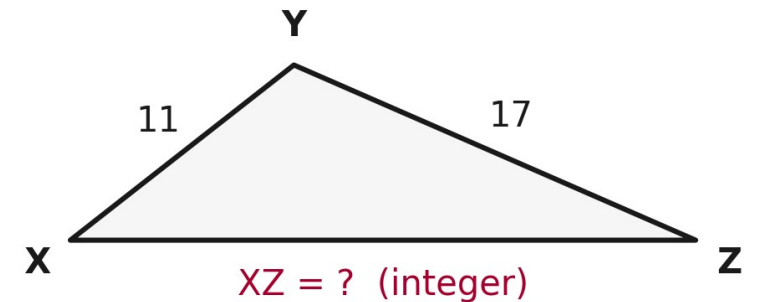
Formula / Rule

Triangle Inequality Theorem: the length of any side must be less than the sum of the other two sides and greater than their positive difference.

Solution

1. Apply the Triangle Inequality Theorem to side XZ using the other two sides, 11 and 17: $|17 - 11| < XZ < 17 + 11$.
2. Simplify the left side: $|17 - 11| = 6$.
3. Simplify the right side: $17 + 11 = 28$.
4. So the inequality is: $6 < XZ < 28$.
5. Since XZ must be an integer strictly between 6 and 28, the possible values are 7, 8, 9, ..., 27.
6. Count these integers: $27 - 7 + 1 = 21$.

✓ **Answer: 21**





Problem 6 — Strategy & Elimination



Tip

The Triangle Inequality always produces a range with the SAME structure: (difference of the two known sides) < unknown side < (sum of the two known sides) — memorize that shape rather than re-deriving it each time.



Trap

A very common error is including the endpoints 6 and 28 as valid values (using $\leq$ instead of $<$), which overcounts the number of possible integer values by 2 — a triangle cannot actually have a side length equal to the sum or difference of the other two, since that would collapse it into a straight line.



Shortcut

Once you have the range $6 < XZ < 28$, the count of integers strictly between two integers a and b is always $(b - a - 1)$ — here, $28 - 6 - 1 = 21$, without needing to list any of the values out.



Desmos Tip

Not a graphing question, but you can verify the count by typing $28 - 6 - 1$ into Desmos to confirm it returns 21.



Process of Elimination

Correct answer: **21**

Tempting wrong answer	Exactly what went wrong
23	Included both endpoints: $6 \leq XZ \leq 28$.
27	Ignored the lower bound: counted 1 through 27.
22	Included one endpoint but not the other.

Problem 7

Very Hard

In right triangle ABC, the right angle is at C. The altitude from C to hypotenuse AB meets AB at point D, dividing it into segments AD = 4 and DB = 9. What is the length of CD?

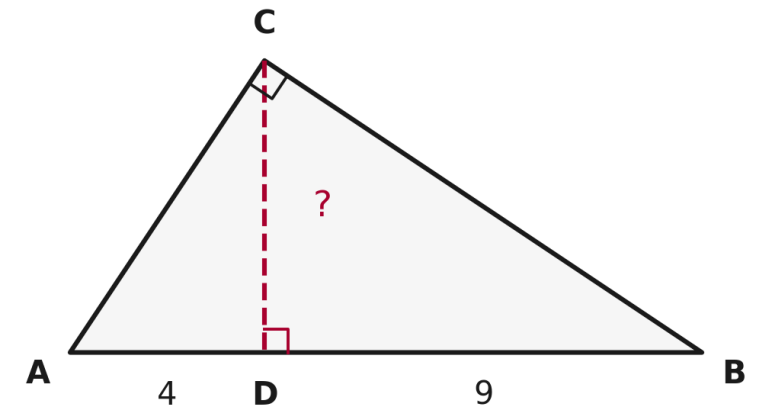
Formula / Rule

Geometric Mean (Altitude-on-Hypotenuse) Relationship: $CD^2 = AD \times DB$.

Solution

1. Drawing the altitude from the right angle to the hypotenuse creates two smaller triangles, both similar to the original triangle and to each other.
2. This similarity gives the relationship $CD^2 = AD \times DB$.
3. Substitute the given values: $CD^2 = 4 \times 9 = 36$.
4. Take the square root of both sides: $CD = 6$ (length must be positive).

 **Answer: 6**





Problem 7 — Strategy & Elimination



Tip

Whenever you see an altitude drawn from a right angle to the hypotenuse, expect a 'geometric mean' relationship — the altitude is the geometric mean of the two segments it creates on the hypotenuse ($CD^2 = AD \times DB$), and each leg is the geometric mean of the whole hypotenuse and its adjacent segment.



Trap

A common error is treating the two hypotenuse segments as if they should simply be averaged (arithmetic mean) rather than multiplied and square-rooted (geometric mean), which produces a different, incorrect value.



Shortcut

Recognize $4 \times 9 = 36$ as a perfect square immediately, so $\sqrt{36} = 6$ requires no calculator — many Digital SAT altitude problems are deliberately built with segment lengths whose product is a perfect square.



Desmos Tip

Type `sqrt(4*9)` into Desmos to instantly confirm $CD = 6$.



Process of Elimination

Correct answer: **6**

Tempting wrong answer	Exactly what went wrong
6.5	Used the arithmetic mean: $(4 + 9)/2$.
13	Found the hypotenuse $AB = 4 + 9$, not the altitude.
36	Stopped at $CD^2 = 36$ — no square root.

Problem 8

Hard

In isosceles triangle DEF, $DE = DF$, and the exterior angle at vertex F measures 130° . What is the measure of the vertex angle at D?

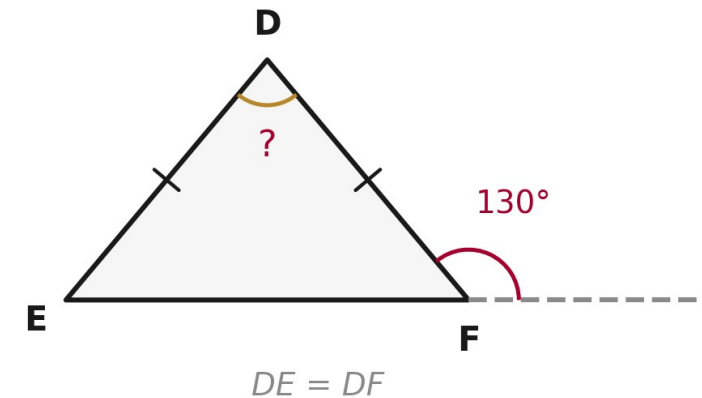
Formula / Rule

Exterior/interior angles are supplementary; Isosceles Triangle Theorem; Triangle Angle Sum Theorem.

Solution

1. The exterior angle at F and the interior angle at F are supplementary (they form a straight line), so: interior angle F = $180 - 130 = 50^\circ$.
2. Since $DE = DF$, the triangle is isosceles with the equal sides being DE and DF — this makes E and F the base vertices, so their opposite (base) angles are equal: angle E = angle F = 50° .
3. Apply the Triangle Angle Sum Theorem: angle D + angle E + angle F = 180.
4. Substitute the known angles: angle D + $50 + 50 = 180$.
5. Simplify: angle D + 100 = 180.
6. Subtract 100 from both sides: angle D = 80.

✓ **Answer:** 80°





Problem 8 — Strategy & Elimination



Tip

When you're given an exterior angle at one of the BASE vertices of an isosceles triangle, convert it to the interior angle first (180° minus the exterior angle) — that interior angle is one of the two equal base angles, which unlocks the rest of the triangle.



Trap

A common error is treating the exterior angle (130°) itself as one of the triangle's interior base angles, skipping the necessary step of subtracting from 180° to find the actual interior angle at F.



Shortcut

Since the exterior angle at a base vertex of an isosceles triangle is supplementary to a base angle, and the vertex angle equals 180° minus twice that base angle, you can combine both steps: vertex angle = $2(\text{exterior angle}) - 180 = 2(130) - 180 = 80$, in one line.



Desmos Tip

Type $2 * 130 - 180$ into Desmos to confirm it returns 80 directly.



Process of Elimination

Correct answer: **80°**

Tempting wrong answer	Exactly what went wrong
50°	Reported the base angle at F ($180^\circ - 130^\circ$), not angle D.
65°	Halved the exterior angle: $130^\circ/2$ — no valid relationship.
100°	Stopped at $E + F = 100^\circ$ — forgot to subtract from 180° .

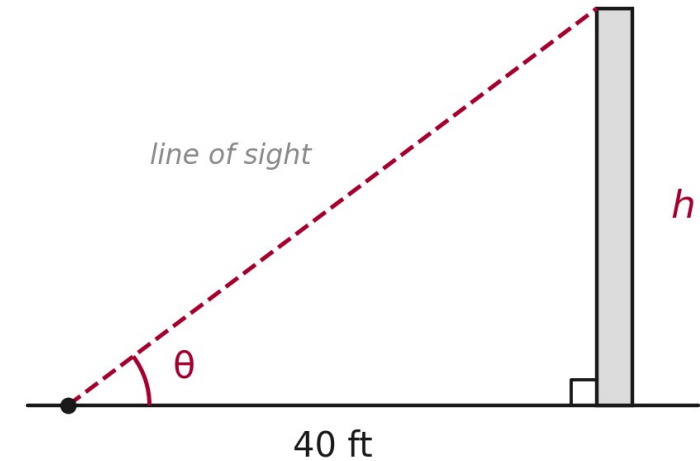
03

Right Triangles and Trigonometry

Pythagoras, SOH-CAH-TOA, complementary angles, and special right triangles.

In this section

- Key terminology — English / Ελληνικά
- Condensed theory with diagrams
- 8 worked problems (Hard / Very Hard)



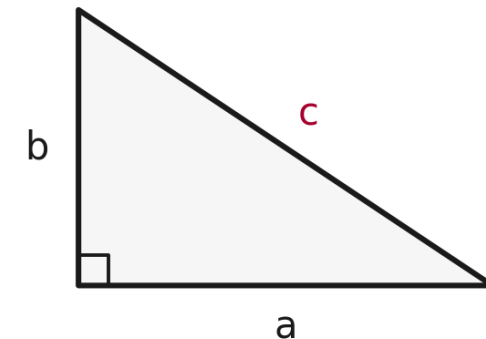
$$\tan \theta = \frac{3}{4}$$

Key Terminology

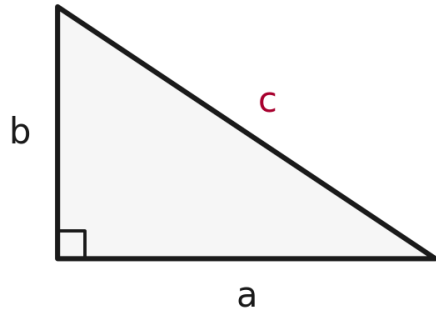
Term (English)	Ελληνικά
Hypotenuse	Υποτείνουσα
Leg (right triangle)	Κάθετη πλευρά
Opposite side	Απέναντι πλευρά
Adjacent side	Προσκειμένη πλευρά
Sine	Ημίτονο
Cosine	Συνημίτονο
Tangent	Εφαπτομένη
Complementary angles	Συμπληρωματικές γωνίες
Pythagorean theorem	Πυθαγόρειο θεώρημα
Angle of elevation	Γωνία ανύψωσης

A Why it matters

- Every question reduces to a handful of relationships.
- The twist is rarely the trig itself.
- It is spotting which side is opposite, adjacent, or hidden in a word problem.



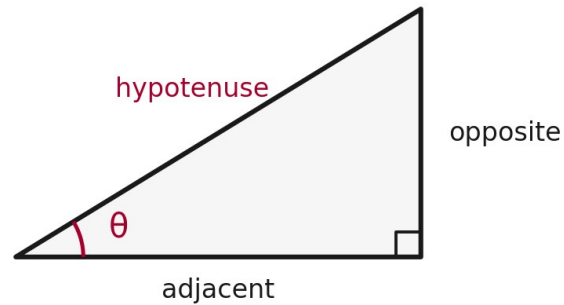
Condensed Theory (1/2)



Pythagorean Theorem

$$a^2 + b^2 = c^2$$

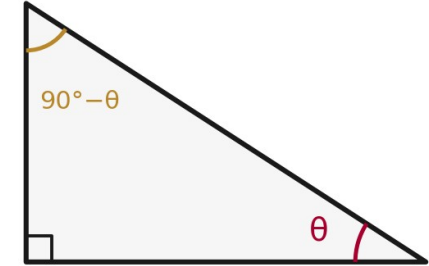
- c is always the hypotenuse — the longest side.
- Right triangles only.



SOH-CAH-TOA

$$\sin = O/H \quad \cos = A/H \quad \tan = O/A$$

- Opposite / adjacent depend on the angle you use.
- Relabel when the angle changes.

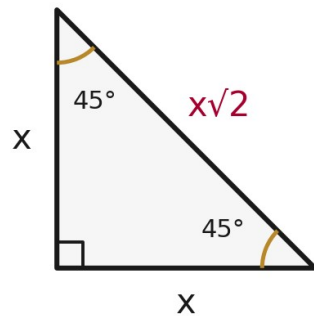


Complementary Angles

$$\sin \theta = \cos(90^\circ - \theta)$$

- The two acute angles sum to 90° .
- Does NOT work for tangent.

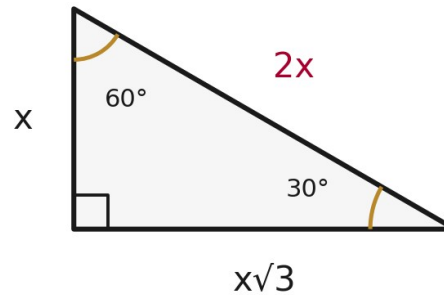
Condensed Theory (2/2)



45-45-90 Triangle

$$x : x : x\sqrt{2}$$

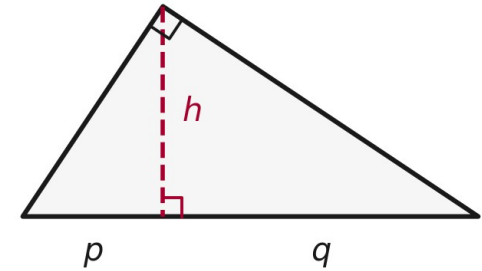
- Half of a square cut on its diagonal.
- Hypotenuse = leg $\times \sqrt{2}$.



30-60-90 Triangle

$$x : x\sqrt{3} : 2x$$

- Half of an equilateral triangle.
- Hypotenuse = 2 $\times$ shortest side.



$$h^2 = p \cdot q$$

Altitude on Hypotenuse

$$h^2 = p \cdot q \quad \text{leg}^2 = (\text{adj. segment}) (\text{whole})$$

- The altitude creates two similar triangles.
- Pick the rule that matches the given length.

Watch Out



Complementary shortcut: sine and cosine only

- $\sin \theta = \cos(90^\circ - \theta)$ is always true.
- Never extend the pattern to tangent.



Label before you calculate

- Find the hypotenuse first.
- Then label opposite / adjacent relative to the angle actually asked about.

Problem 1

Hard

In right triangle ABC, angle C = 90° , AC = 9, and BC = 12. What is $\sin(A)$?

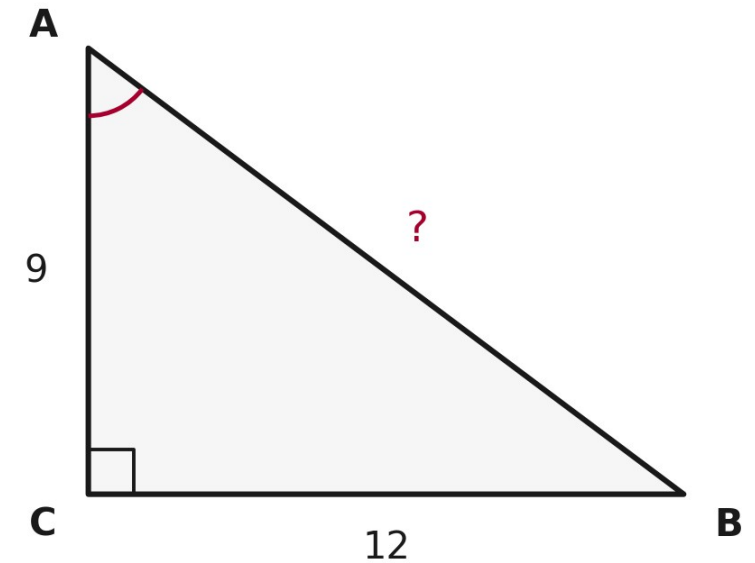
Formula / Rule

Pythagorean theorem to find the hypotenuse, then SOH-CAH-TOA.

Solution

1. Since angle C = 90° , side AB is the hypotenuse. Find it using the Pythagorean theorem: $AC^2 + BC^2 = AB^2$.
2. Substitute: $9^2 + 12^2 = AB^2$, so $81 + 144 = 225 = AB^2$.
3. Take the square root: $AB = 15$.
4. For angle A, the side opposite is BC (the side that doesn't touch vertex A), and the hypotenuse is AB.
5. $\sin(A) = \text{opposite/hypotenuse} = BC/AB = 12/15$.
6. Simplify the fraction: $12/15 = 4/5$.

 **Answer:** $4/5$ (or 0.8)





Problem 1 — Strategy & Elimination



Tip

Before computing any trig ratio, find the missing side with the Pythagorean theorem first if it isn't given — you can't apply SOH-CAH-TOA correctly with only two of the three sides.



Shortcut

Recognize 9-12-15 as a multiple of the 3-4-5 Pythagorean triple (three times 3-4-5) the moment you see two of the three sides — this lets you find $AB = 15$ by inspection instead of computing the full Pythagorean theorem by hand.



Trap

A common error is using AC (which is adjacent to angle A , not opposite it) in the numerator instead of BC , confusing which leg is 'opposite' angle A .



Desmos Tip

Type $\text{sqrt}(9^2+12^2)$ into Desmos to confirm the hypotenuse is 15, then $12/15$ to confirm $\sin(A) = 0.8$.



Process of Elimination

Correct answer: **4/5 (or 0.8)**

Tempting wrong answer	Exactly what went wrong
3/5	Used AC (adjacent to A) instead of BC : $9/15$.
4/3	Computed $\tan A = 12/9$ instead of $\sin A$.
5/4	Inverted the ratio: hypotenuse/opposite.

Problem 2

Hard

In right triangle DEF, angle E = 90° , angle D = 30° , and the hypotenuse DF = 14. What is the perimeter of triangle DEF?

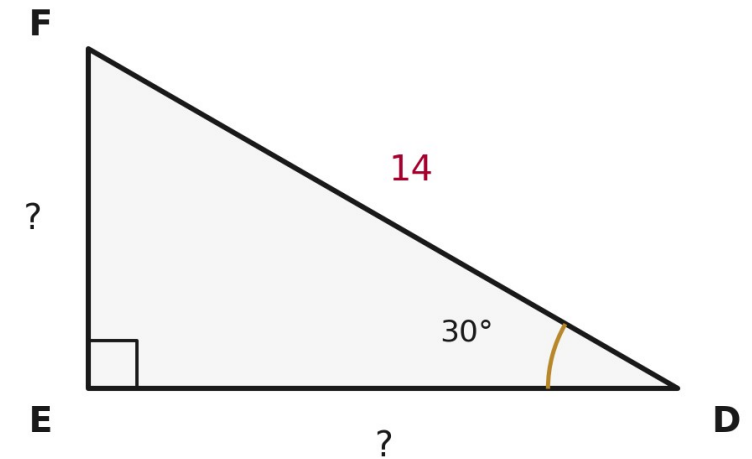
Formula / Rule

30-60-90 special right triangle side ratios: x , $x\sqrt{3}$, $2x$ (opposite 30° , 60° , 90° respectively).

Solution

1. Since angle E = 90° and angle D = 30° , the third angle, angle F, must be 60° (angles sum to 180°).
2. In a 30-60-90 triangle, the hypotenuse (opposite the 90° angle) is always twice the shortest side (opposite the 30° angle).
3. Since the hypotenuse DF = 14, the shortest side (EF, opposite the 30° angle at D) equals $14 \div 2 = 7$.
4. The remaining side (DE, opposite the 60° angle) equals the shortest side times $\sqrt{3}$: $DE = 7\sqrt{3}$.
5. Find the perimeter by adding all three sides: $14 + 7 + 7\sqrt{3}$.
6. Combine the whole-number terms: perimeter = $21 + 7\sqrt{3}$.

✓ **Answer:** $21 + 7\sqrt{3}$





Problem 2 — Strategy & Elimination



Tip

In any 30-60-90 triangle, identify the side opposite the 30° angle first — it's always the shortest side and the 'base unit' (x) that the other two sides ($x\sqrt{3}$ and $2x$) are built from.



Shortcut

Memorize the 30-60-90 ratio as a package ($x : x\sqrt{3} : 2x$) so that once you identify ANY one side, you can write down the other two immediately — here, hypotenuse 14 means $x = 7$ instantly, without solving an equation.



Trap

A common error is assuming the given hypotenuse (14) IS the base unit x , then computing the other sides as $14\sqrt{3}$ and 28 instead of first dividing by 2 to find the correct base unit of 7.



Desmos Tip

Type $14 + 7 + 7 * \text{sqrt}(3)$ into Desmos to get the decimal value ≈ 33.12 as a check on your exact-form answer.



Process of Elimination

Correct answer: $21 + 7\sqrt{3}$

Tempting wrong answer	Exactly what went wrong
$42 + 14\sqrt{3}$	Treated the hypotenuse 14 as x : sides 14, $14\sqrt{3}$, 28.
$21 + 14\sqrt{3}$	Found $x = 7$ but wrote the long leg as $14\sqrt{3}$.
$7 + 7\sqrt{3}$	Added only the two legs — left out the hypotenuse.

Problem 3

Hard

In right triangle ABC, angle C = 90° . If $\sin(A) = \cos(3x + 6)^\circ$ and angle A = $(x + 24)^\circ$, what is the value of x?

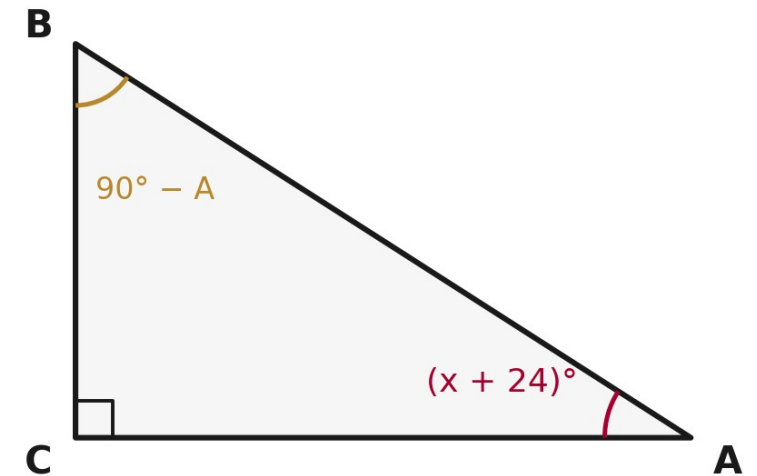
Formula / Rule

Complementary Angle Relationship: $\sin(\theta) = \cos(90^\circ - \theta)$.

Solution

1. Since $\sin(A) = \cos(3x+6)^\circ$, the complementary angle relationship tells us that $(3x+6)$ must equal 90° minus angle A: $3x + 6 = 90 - (x + 24)$.
2. Simplify the right side: $90 - (x + 24) = 66 - x$.
3. Rewrite the equation: $3x + 6 = 66 - x$.
4. Add x to both sides: $4x + 6 = 66$.
5. Subtract 6 from both sides: $4x = 60$.
6. Divide by 4: $x = 15$.

✓ **Answer: 15**



$$\sin A = \cos (3x + 6)^\circ$$



Problem 3 — Strategy & Elimination



Tip

Whenever you see $\sin$ of one angle set equal to $\cos$ of another expression, immediately translate it using $\sin(\theta) = \cos(90^\circ - \theta)$ — this converts the trig equation into a purely algebraic one about the angle measures.



Trap

A common error is setting the two angle expressions themselves equal to each other ($3x+6 = x+24$) instead of setting one equal to 90° MINUS the other, forgetting that sine and cosine of the SAME angle aren't generally equal — only complementary angles' sine and cosine match.



Shortcut

Since $\sin(A) = \cos(90-A)$ always, and here $\sin(A) = \cos(3x+6)$, you can immediately write $3x+6 = 90-A$ and substitute $A = x+24$ directly, without introducing any extra variables.



Desmos Tip

Type $\sin(15+24)$ and $\cos(3 \cdot 15+6)$ into Desmos (in degree mode) to confirm both evaluate to the same value, verifying $x = 15$.



Process of Elimination

Correct answer: 15

Tempting wrong answer	Exactly what went wrong
9	Set the angles equal: $3x + 6 = x + 24$.
27	Sign error: $90 - (x + 24)$ written as $114 - x$.
39	Found $x = 15$ but reported angle $A = x + 24$.

Problem 4

Very Hard

In right triangle ABC, angle C = 90°, and $\tan(A) = 5/12$. What is the value of $\sin(A) + \cos(A)$?

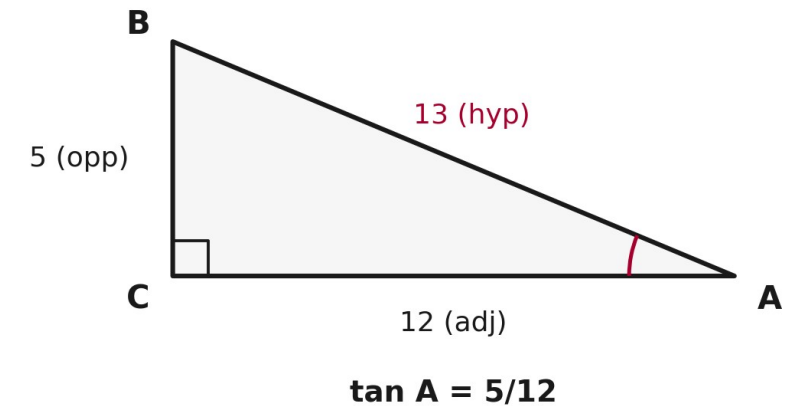
Formula / Rule

SOH-CAH-TOA, using a Pythagorean triple implied by the given tangent ratio.

Solution

1. $\tan(A) = \text{opposite}/\text{adjacent} = 5/12$, so let the side opposite angle A have length 5 and the side adjacent to angle A have length 12 (using the simplest such triangle).
2. Find the hypotenuse using the Pythagorean theorem: $5^2 + 12^2 = \text{hyp}^2$, so $25 + 144 = 169 = \text{hyp}^2$.
3. Take the square root: hypotenuse = 13.
4. Find $\sin(A) = \text{opposite}/\text{hypotenuse} = 5/13$.
5. Find $\cos(A) = \text{adjacent}/\text{hypotenuse} = 12/13$.
6. Add them together: $\sin(A) + \cos(A) = 5/13 + 12/13 = 17/13$.

 **Answer:** 17/13





Problem 4 — Strategy & Elimination



Tip

When you're given only a tangent ratio and need sine or cosine, build the actual right triangle from the ratio (opposite = numerator, adjacent = denominator) and use the Pythagorean theorem to find the hypotenuse — don't try to convert tan directly to sin or cos with a formula.



Trap

A common error is assuming $\tan(A) = 5/12$ directly gives $\sin(A) = 5$ and $\cos(A) = 12$ (treating the tangent's numerator and denominator as if they were already the sine and cosine values), skipping the hypotenuse calculation entirely.



Shortcut

Recognize 5-12-13 as a classic Pythagorean triple immediately from the given tangent ratio $5/12$ — this lets you skip the Pythagorean theorem calculation and write down the hypotenuse (13) by memory.



Desmos Tip

Type $5/13 + 12/13$ into Desmos to confirm it simplifies to $17/13$ (≈ 1.31).



Process of Elimination

Correct answer: **17/13**

Tempting wrong answer	Exactly what went wrong
17	Added $5 + 12$ — treated tan's parts as sin and cos.
1	Confused $\sin A + \cos A$ with $\sin^2 A + \cos^2 A = 1$.
7/13	Computed $\cos A - \sin A$ instead of the sum.

Problem 5

Hard

A square has a diagonal of length $10\sqrt{2}$. What is the area of the square?

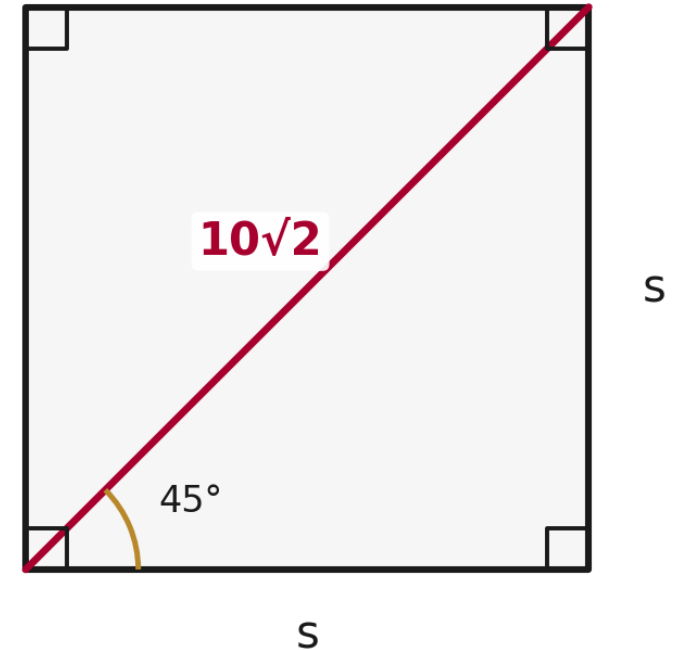
Formula / Rule

45-45-90 triangle relationship: a square's diagonal splits it into two 45-45-90 triangles, where the diagonal (hypotenuse) equals side $\times \sqrt{2}$.

Solution

1. A square's diagonal divides it into two 45-45-90 right triangles, where each leg is a side of the square and the hypotenuse is the diagonal.
2. Using the 45-45-90 relationship, diagonal = side $\times \sqrt{2}$.
3. Substitute the given diagonal: $10\sqrt{2} = \text{side} \times \sqrt{2}$.
4. Divide both sides by $\sqrt{2}$: side = 10.
5. Find the area of the square: Area = side² = $10^2 = 100$.

✓ **Answer:** 100





Problem 5 — Strategy & Elimination



Tip

Whenever a square's diagonal is given in the form $(\text{number})\sqrt{2}$, the side length is simply that same number — the $\sqrt{2}$ factor from the 45-45-90 relationship cancels out cleanly.



Shortcut

Memorize that for a square, $\text{diagonal} = \text{side} \times \sqrt{2}$, so $\text{area} = \text{side}^2 = (\text{diagonal}/\sqrt{2})^2 = \text{diagonal}^2/2$ — here, $(10\sqrt{2})^2/2 = 200/2 = 100$, in one direct computation without isolating the side length first.



Trap

A common error is treating the given diagonal $(10\sqrt{2})$ as if it were the side length itself and squaring it directly, which produces an area many times too large.



Desmos Tip

Type $(10 * \text{sqrt}(2))^2 / 2$ into Desmos to directly confirm the area is 100.



Process of Elimination

Correct answer: **100**

Tempting wrong answer	Exactly what went wrong
200	Squared the diagonal: $(10\sqrt{2})^2$ — treated it as the side.
50	Divided the diagonal by 2 instead of $\sqrt{2}$: side $5\sqrt{2}$.
≈ 14.1	Reported the diagonal's decimal value, not the area.

Problem 6

Very Hard

In right triangle ABC, angle C = 90° , and the altitude from C meets hypotenuse AB at point D. If AD = 8 and leg AC = 12, what is the length of DB?

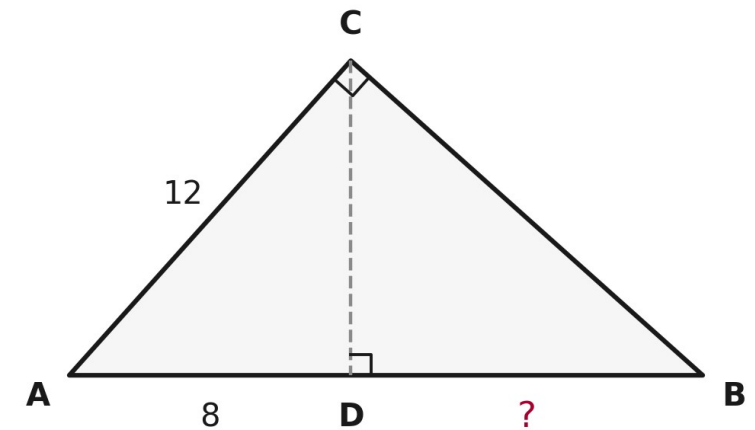
Formula / Rule

Altitude-on-Hypotenuse relationship: $\text{leg}^2 = (\text{adjacent hypotenuse segment})(\text{whole hypotenuse})$.

Solution

1. The leg AC is adjacent to segment AD on the hypotenuse, so the relevant relationship is: $AC^2 = AD \times AB$ (where AB is the FULL hypotenuse, not just one segment).
2. Substitute the known values: $12^2 = 8 \times AB$.
3. Simplify: $144 = 8 \times AB$.
4. Divide both sides by 8: $AB = 18$.
5. Since AB is made up of segments AD and DB, and $AD = 8$: $DB = AB - AD = 18 - 8 = 10$.

 **Answer: 10**





Problem 6 — Strategy & Elimination



Tip

For altitude-on-hypotenuse problems, there are two related but different relationships — $\text{altitude}^2 = (\text{segment})(\text{segment})$ for the altitude itself, and $\text{leg}^2 = (\text{adjacent segment})(\text{whole hypotenuse})$ for either leg. Identify which one the given information matches before setting up an equation.



Trap

A common error is applying the altitude-only relationship ($CD^2 = AD \times DB$) here, when the problem actually gives a LEG length (AC), which requires the other relationship ($\text{leg}^2 = \text{adjacent segment} \times \text{whole hypotenuse}$) instead.



Shortcut

Since $12^2 = 144$ and $144 \div 8 = 18$ is straightforward mental math, you can solve for the full hypotenuse AB in one step and then just subtract the given AD, without introducing extra variables for DB.



Desmos Tip

Type $12^2/8$ into Desmos to confirm $AB = 18$, then $18 - 8$ to confirm $DB = 10$.



Process of Elimination

Correct answer: 10

Tempting wrong answer	Exactly what went wrong
18	Found $AB = 18$ and stopped (or misused $AC^2 = AD \cdot DB$).
$4\sqrt{5}$	Found the altitude $CD = \sqrt{(12^2 - 8^2)}$ instead of DB.
4	Subtracted lengths: $12 - 8$.

Problem 7

Very Hard

From a point on the ground 40 feet from the base of a tower, the angle of elevation to the top of the tower is θ , where $\tan(\theta) = 3/4$. What is the height of the tower, in feet?

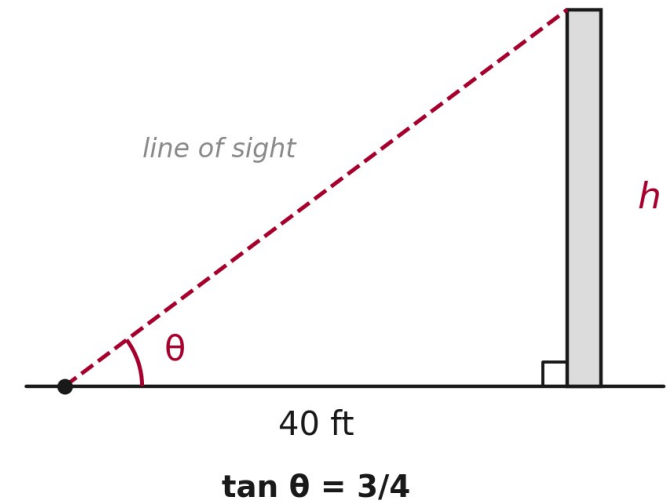
Formula / Rule

SOH-CAH-TOA (tangent) applied to a right triangle formed by the tower, the ground, and the line of sight.

Solution

1. The tower's height, the 40-foot ground distance, and the line of sight form a right triangle, with the right angle at the base of the tower.
2. The angle of elevation θ is at the point on the ground, so the tower's height is the side OPPOSITE θ , and the 40-foot distance is the side ADJACENT to θ .
3. Apply the tangent ratio: $\tan(\theta) = \text{opposite/adjacent} = \text{height}/40$.
4. Substitute the given value: $3/4 = \text{height}/40$.
5. Cross-multiply (or multiply both sides by 40): $\text{height} = 40 \times (3/4)$.
6. Simplify: $\text{height} = 30$.

✓ **Answer:** 30 feet





Problem 7 — Strategy & Elimination



Tip

For any 'angle of elevation' word problem, first identify the right triangle hidden in the description: the vertical height, the horizontal ground distance, and the line of sight are always the two legs and the hypotenuse (or, if only two of the three are relevant, just the two legs).



Trap

A common error is applying the given ratio (3/4) as sine or cosine instead of tangent, which would require knowing the hypotenuse (the line of sight distance) — information this problem never gives.



Shortcut

Recognize $\tan(\theta) = 3/4$ as implying a 3-4-5 right triangle by ratio — so height:40 is in the ratio 3:4, meaning height = $(3/4)(40) = 30$ directly, the same computation as scaling a 3-4-5 triangle up by a factor of 10.



Desmos Tip

Type $40 \cdot (3/4)$ into Desmos to directly confirm the height is 30.



Process of Elimination

Correct answer: **30 feet**

Tempting wrong answer	Exactly what went wrong
≈ 53.3	Inverted the ratio: $40 \times 4/3$.
50	Found the line of sight (a 30-40-50 triangle), not the height.
40	Reported the ground distance.

Problem 8

Hard

In right triangle ABC, angle C = 90° . If $\sin(B) = 7/25$, what is $\cos(A)$?



Formula / Rule

Complementary Angle Relationship: in a right triangle, the two non-right angles are complementary, so $\sin$ of one equals $\cos$ of the other.

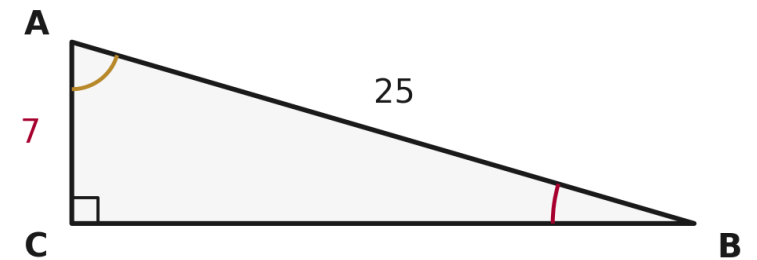


Solution

1. Since angle C = 90° , angles A and B are the two remaining angles in the triangle, and they must be complementary: $A + B = 90^\circ$.
2. Because A and B are complementary, $\cos(A) = \cos(90^\circ - B) = \sin(B)$.
3. Substitute the given value: $\cos(A) = \sin(B) = 7/25$.



Answer: $7/25$



$$\sin B = 7/25 \rightarrow \cos A = ?$$



Problem 8 — Strategy & Elimination



Tip

Whenever a right-triangle problem gives you the sine of one acute angle and asks for the cosine of the OTHER acute angle, the answer is immediately the same value — no triangle needs to be drawn or solved.



Trap

A common error is trying to find the third side of the triangle using the Pythagorean theorem (treating 7 and 25 as two actual side lengths) and then computing $\cos(A)$ from scratch, when the complementary relationship gives the answer directly without any extra computation.



Shortcut

The relationship $\cos(A) = \sin(B)$ for complementary angles A and B means you can answer this type of question by inspection the moment you see 'angle C = 90°' and a sine or cosine value for one of the other two angles — no computation required at all.



Desmos Tip

Not needed here since no triangle needs to be solved, but you could type $\sin(7/25)$ and 90-that into Desmos, then take the cosine, to confirm it matches 7/25 numerically.



Process of Elimination

Correct answer: 7/25

Tempting wrong answer	Exactly what went wrong
24/25	Treated 7 and 25 as sides, found 24, then used it for the wrong angle.
25/7	Inverted the value: used $1/\sin B$.
18/25	Subtracted: $(25 - 7)/25$.

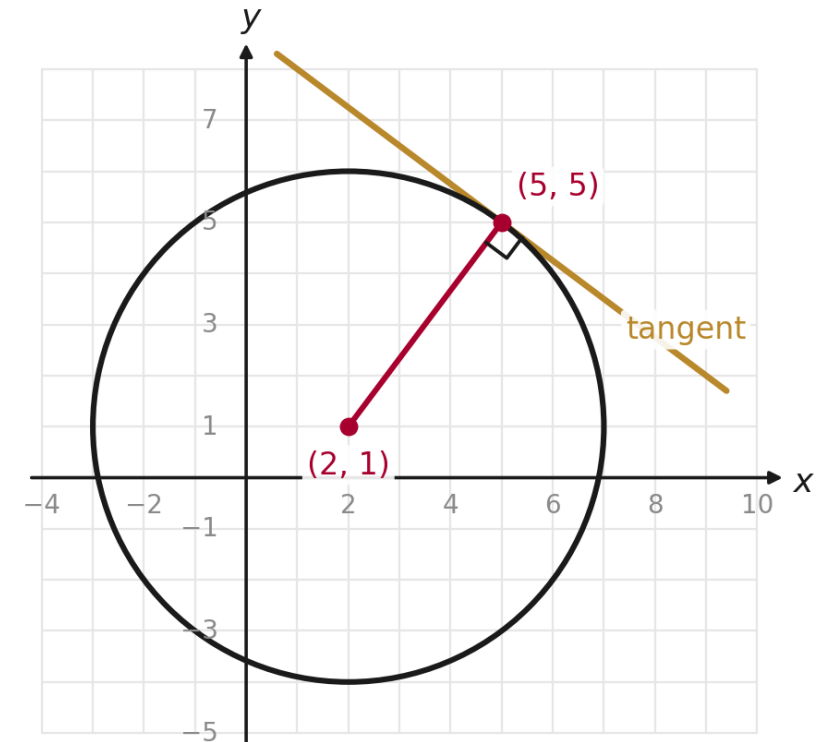
04

Circles

Circle equations, arcs, sectors, radians, inscribed angles, and tangents.

In this section

- Key terminology — English / Ελληνικά
- Condensed theory with diagrams
- 8 worked problems (Hard / Very Hard)

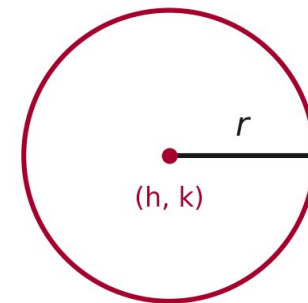


Key Terminology

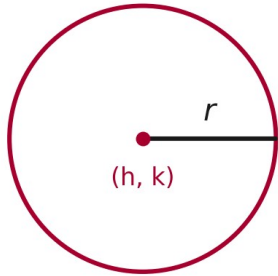
Term (English)	Ελληνικά
Center	Κέντρο
Radius	Ακτίνα
Diameter	Διάμετρος
Chord	Χορδή
Tangent line	Εφαπτομένη
Arc	Τόξο
Central angle	Επίκεντρη γωνία
Inscribed angle	Εγγεγραμμένη γωνία
Sector	Κυκλικός τομέας
Radian	Ακτίνιο

A **α** Why it matters

- Algebraic family: the circle equation (usually completing the square).
- Geometric family: arcs, sectors, angles, tangents.
- Recognize which relationship the figure or equation is testing.



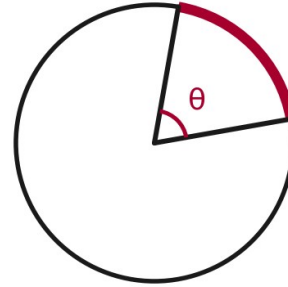
Condensed Theory (1/2)



Standard Form

$$(x - h)^2 + (y - k)^2 = r^2$$

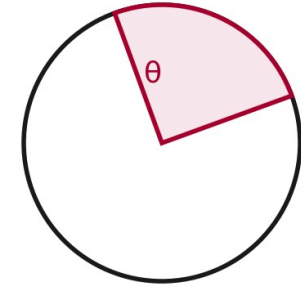
- Center (h, k), radius r.
- Expanded form? Complete the square on x and y.



Arc Length

$$\text{arc} = (\theta/360) \cdot 2\pi r$$

- θ in degrees.
- An arc is a fraction of the circumference.

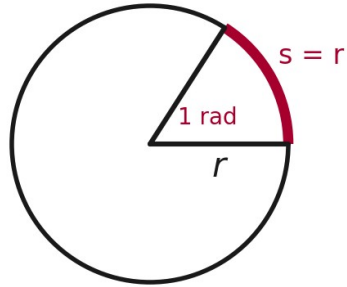


Sector Area

$$A = (\theta/360) \cdot \pi r^2$$

- θ in degrees.
- Same fraction, applied to the full area.

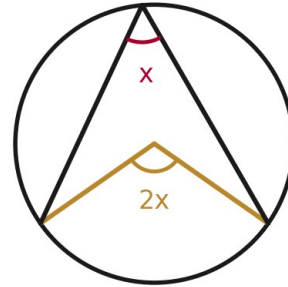
Condensed Theory (2/2)



Radians

$$\text{rad} = \text{deg} \times \pi/180 \quad s = r\theta$$

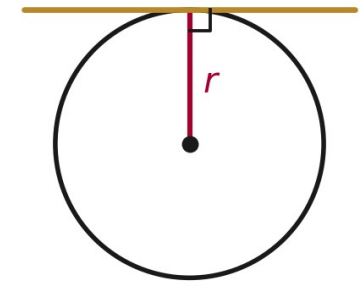
- In radians, arc length = radius $\times$ angle.
- No 360 fraction needed.



Inscribed Angle Theorem

$$\text{inscribed} = \frac{1}{2} \cdot \text{central}$$

- Both angles must subtend the same arc.
- Never set them equal.



Tangent Line

$$\text{tangent} \perp \text{radius}$$

- Perpendicular at the point of tangency.
- Tangent slope = negative reciprocal of radius slope.

Watch Out



The right side is r^2 , not r

- After completing the square, take the square root as your LAST step.



Thales' Theorem

- A triangle inscribed with a diameter as one side has a 90° angle opposite that diameter.
- The test rarely states it — you must spot it.

Problem 1

Hard

A circle is defined by the equation $x^2 + y^2 - 6x + 8y - 11 = 0$. What is the radius of the circle?

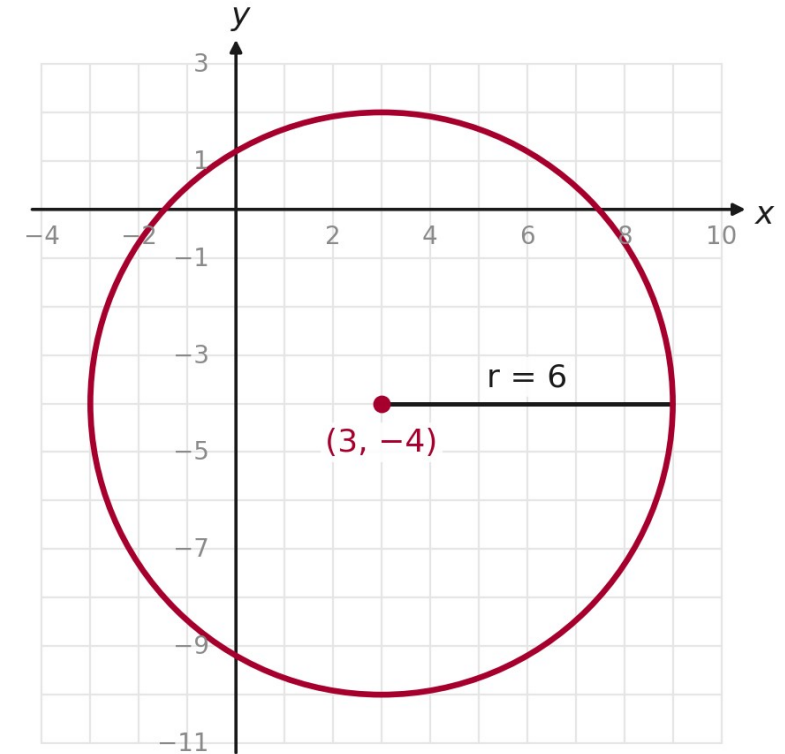
Formula / Rule

Complete the square on both x and y to convert to standard form $(x-h)^2 + (y-k)^2 = r^2$.

Solution

1. Group the x -terms and y -terms, and move the constant to the other side: $(x^2 - 6x) + (y^2 + 8y) = 11$.
2. Complete the square for the x -terms: take half of -6 (which is -3), square it (9), and add it to both sides: $(x^2 - 6x + 9)$.
3. Complete the square for the y -terms: take half of 8 (which is 4), square it (16), and add it to both sides: $(y^2 + 8y + 16)$.
4. Add both 9 and 16 to the right side as well: $(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16$.
5. Rewrite as perfect squares: $(x - 3)^2 + (y + 4)^2 = 36$.
6. Since $r^2 = 36$, take the square root: $r = 6$.

✓ **Answer: 6**



$$x^2 + y^2 - 6x + 8y - 11 = 0$$



Problem 1 — Strategy & Elimination



Tip

When completing the square for a circle equation, always add the SAME amount you add to the left side to the right side too — it's easy to complete the square correctly on the left but forget to balance the equation on the right.



Shortcut

Since the coefficient of x and y is always 1 in these problems, the number you add for each variable is always (half of that variable's linear coefficient), squared — for $-6x$, that's $(-3)^2=9$; for $+8y$, that's $(4)^2=16$. Recognizing this pattern lets you complete the square without rewriting the grouping step explicitly.



Trap

A very common error is reporting the value of r^2 (36) as the radius itself, forgetting the final square-root step.



Desmos Tip

Type $(x-3)^2+(y+4)^2=36$ into Desmos and confirm it draws a circle whose radius (readable from the graph) is 6.



Process of Elimination

Correct answer: 6

Tempting wrong answer	Exactly what went wrong
36	Reported r^2 as the radius.
3	Reported the center's x -coordinate.
$\sqrt{14}$	Sign error on the constant: $-11 + 9 + 16 = 14$.

Problem 2

Hard

A circle has radius 9 centimeters. What is the length of an arc with a central angle of 40° , in terms of π ?



Formula / Rule

Arc length = $(\theta/360)(2\pi r)$, with θ in degrees.

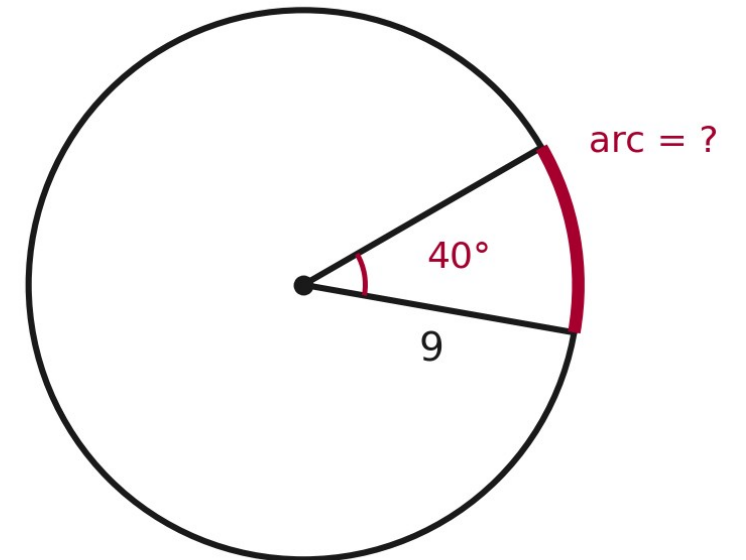


Solution

1. Substitute the given values into the arc length formula: arc length = $(40/360)(2\pi)(9)$.
2. Simplify the fraction $40/360 = 1/9$.
3. Compute $2\pi(9) = 18\pi$.
4. Multiply: $(1/9)(18\pi) = 2\pi$.



Answer: 2π centimeters





Problem 2 — Strategy & Elimination



Tip

Always simplify the fraction ($\theta/360$) to its lowest terms first — it often cancels neatly with the circumference, as it does here ($40/360$ reduces to $1/9$, matching the radius 9 nicely).



Shortcut

For circles where the radius is a nice divisor of the corresponding circumference fraction's denominator, the answer often simplifies to a small whole-number multiple of π — a quick sanity check is that an arc's length in terms of π should always be smaller than the full circumference (18π here), which 2π clearly is.



Trap

A common error is stopping at the full circumference (18π), or using the area formula πr^2 instead of the circumference $2\pi r$.



Desmos Tip

Type $(40/360) * 2 * \pi * 9$ into Desmos to confirm it evaluates to 2π (≈ 6.28).



Process of Elimination

Correct answer: **2π centimeters**

Tempting wrong answer	Exactly what went wrong
π	Simplified $40/360$ as $1/18$.
4π	Used the diameter 18 in place of r : $(1/9)(2\pi)(18)$.
18π	Found the full circumference — forgot the $40/360$ fraction.

Problem 3

Hard

A circular pizza has a radius of 8 inches. What is the area of a slice with a central angle of 45° , in terms of π ?



Formula / Rule

Sector area = $(\theta/360)(\pi r^2)$, with θ in degrees.

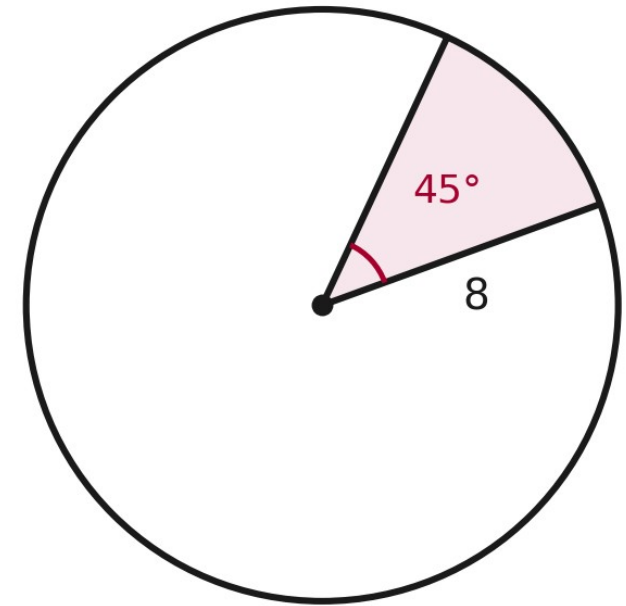


Solution

1. Substitute the given values into the sector area formula: sector area = $(45/360)(\pi)(8)^2$.
2. Simplify the fraction $45/360 = 1/8$.
3. Compute $\pi(8)^2 = 64\pi$.
4. Multiply: $(1/8)(64\pi) = 8\pi$.



Answer: 8π square inches



slice area = ?



Problem 3 — Strategy & Elimination



Tip

Simplify the angle fraction ($\theta/360$) before multiplying by the full circle's area — it's much easier to multiply a simplified fraction like $1/8$ by 64π than to multiply $45/360$ by 64π directly.



Trap

A common error is squaring the radius incorrectly or forgetting to square it at all, using 8 instead of 64 in the sector area formula.



Shortcut

Since 45° is exactly $1/8$ of a full 360° circle, you can immediately write the sector area as $(1/8)$ of the full circle's area — recognizing common angle fractions ($45^\circ=1/8$, $60^\circ=1/6$, $90^\circ=1/4$, $120^\circ=1/3$) by sight saves the division step.



Desmos Tip

Type $(45/360)*\pi*8^2$ into Desmos to confirm it evaluates to 8π (≈ 25.13).



Process of Elimination

Correct answer: **8π square inches**

Tempting wrong answer	Exactly what went wrong
π	Forgot to square the radius: $(1/8)(8\pi)$.
64π	Found the area of the whole pizza.
16π	Simplified $45/360$ as $1/4$.

Problem 4

Hard

A circle has radius 10. An arc on the circle has a central angle of $2\pi/5$ radians. What is the length of the arc?



Formula / Rule

In radians, arc length $s = r\theta$ directly — no fraction of 360 needed.

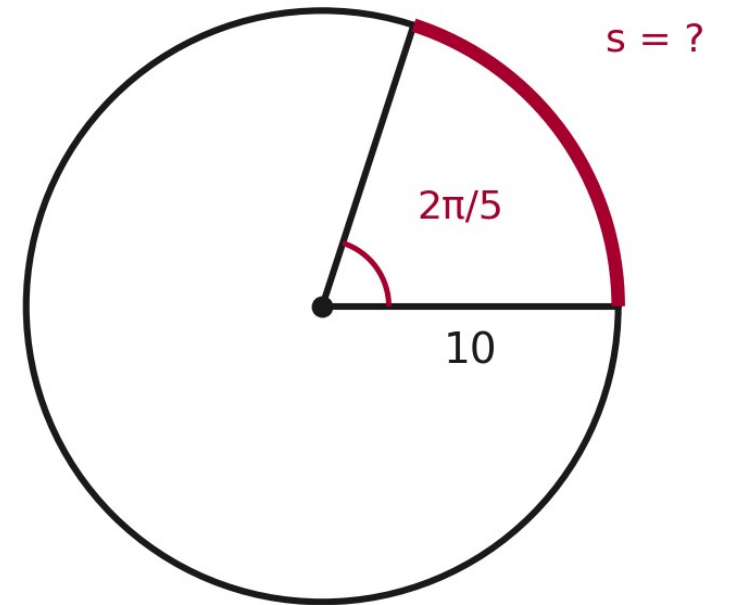


Solution

1. Since the angle is already given in radians, use the simplified arc length formula: $s = r\theta$.
2. Substitute the given values: $s = (10)(2\pi/5)$.
3. Multiply: $10 \times (2\pi/5) = 20\pi/5$.
4. Simplify the fraction: $20\pi/5 = 4\pi$.



Answer: 4π





Problem 4 — Strategy & Elimination



Tip

The moment an angle is given in radians rather than degrees, switch to the simpler formula $s = r\theta$ — there's no need for the $(\theta/360)$ fraction at all, since radians already measure angle as a fraction of the full circle scaled by 2π .



Trap

A common error is still applying the degree-based formula $(\theta/360)(2\pi r)$ with the radian value plugged in for θ , which produces an entirely incorrect result since the formulas aren't interchangeable without converting units first.



Shortcut

Since $s = r\theta$ is direct multiplication with no extra fraction, radian-based arc length problems are usually faster than degree-based ones — just multiply radius by the angle in radians and simplify.



Desmos Tip

Type $10 \cdot (2 \cdot \pi / 5)$ into Desmos to confirm it evaluates to 4π (≈ 12.57).



Process of Elimination

Correct answer: **4π**

Tempting wrong answer	Exactly what went wrong
8π	Used the diameter: $20 \times 2\pi/5$.
4	Dropped the π : $10 \times 2/5$.
720	Converted to 72° , then used $s = r\theta$ with degrees: 10×72 .

Problem 5

Hard

In a circle, a central angle and inscribed angle ABC both intercept the same arc AC (with the central angle's vertex at the circle's center, and the inscribed angle's vertex at point B on the circle). If the central angle measures $(4x + 10)^\circ$ and the inscribed angle ABC measures $(x + 25)^\circ$, what is the measure of the inscribed angle?

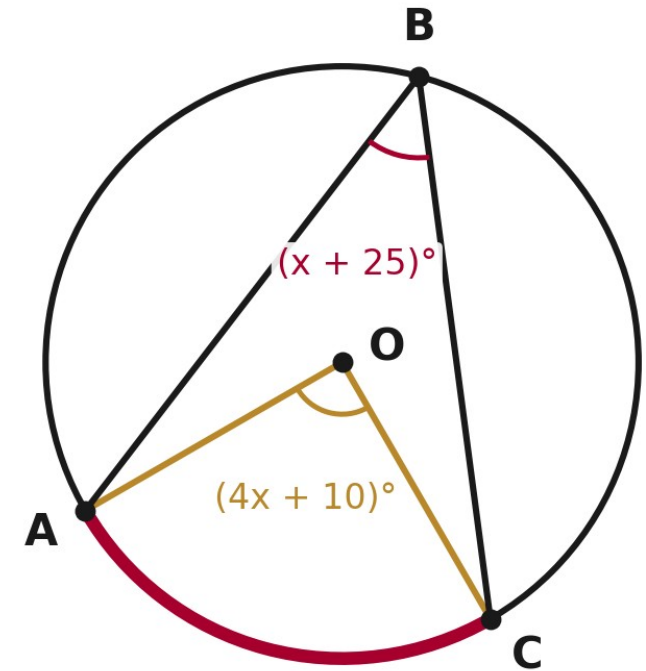
Formula / Rule

Inscribed Angle Theorem: inscribed angle = $(1/2)$ (central angle), for angles subtending the same arc.

Solution

1. Apply the Inscribed Angle Theorem: $(x + 25) = (1/2)(4x + 10)$.
2. Distribute the $1/2$ on the right side: $(1/2)(4x+10) = 2x + 5$.
3. Rewrite the equation: $x + 25 = 2x + 5$.
4. Subtract x from both sides: $25 = x + 5$.
5. Subtract 5 from both sides: $x = 20$.
6. Substitute into the inscribed angle expression: $(20) + 25 = 45$.

✓ **Answer:** 45°





Problem 5 — Strategy & Elimination



Tip

When a problem gives you both a central angle and an inscribed angle subtending the same arc, immediately write 'inscribed = half of central' as your equation — it's a one-line relationship, not something requiring the full circle's angle sum.



Shortcut

Since the inscribed angle is always exactly half the central angle for the same arc, you can substitute directly: $(x+25) = (4x+10)/2$, then multiply both sides by 2 immediately ($2x+50 = 4x+10$) to clear the fraction in one step rather than distributing it first.



Trap

A common error is setting the central angle and the inscribed angle equal to each other, instead of setting the inscribed angle equal to HALF of the central angle.



Desmos Tip

Type $(4 \cdot 20 + 10) / 2$ into Desmos to confirm it equals 45, matching $(20 + 25)$.



Process of Elimination

Correct answer: 45°

Tempting wrong answer	Exactly what went wrong
30°	Set the angles equal: $x + 25 = 4x + 10 \rightarrow x = 5$.
90°	Reported the CENTRAL angle: $4(20) + 10$.
20°	Reported x instead of the angle.

Problem 6

Very Hard

A circle has equation $(x - 2)^2 + (y - 1)^2 = 25$. What is the slope of the line tangent to the circle at the point $(5, 5)$?

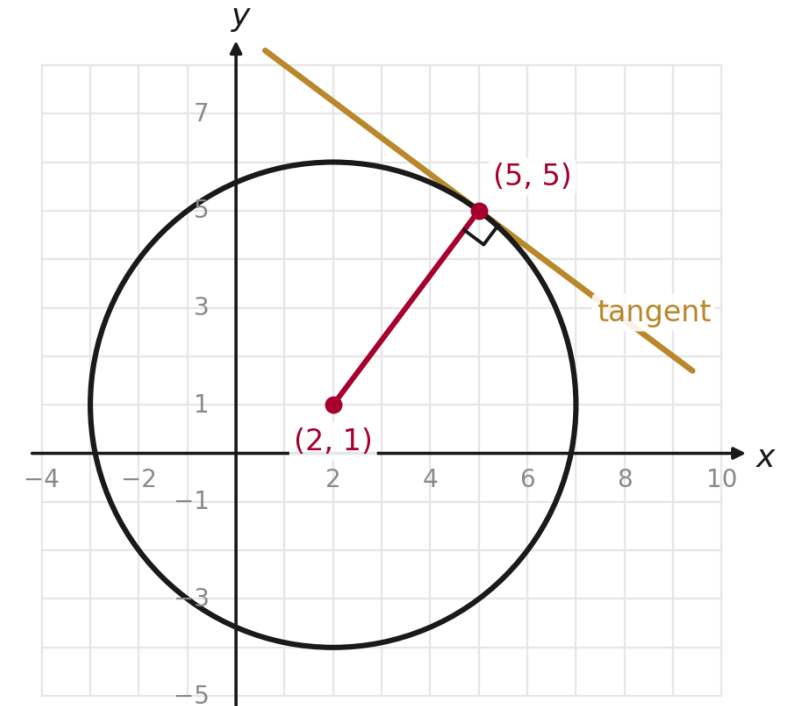
Formula / Rule

A tangent line is perpendicular to the radius drawn to the point of tangency.

Solution

1. Identify the circle's center from the standard form equation: center = $(2, 1)$.
2. Find the slope of the radius connecting the center $(2, 1)$ to the point of tangency $(5, 5)$:
slope = $(5 - 1)/(5 - 2) = 4/3$.
3. Since the tangent line is perpendicular to this radius, its slope is the negative reciprocal of $4/3$.
4. The negative reciprocal of $4/3$ is $-3/4$.

✓ **Answer:** $-3/4$





Problem 6 — Strategy & Elimination



Tip

For any tangent-line-to-a-circle problem, the two-step process is always the same: find the slope of the radius to the given point, then take the negative reciprocal for the tangent line's slope.



Trap

A common error is computing the slope of the radius correctly but then forgetting to take the negative reciprocal, reporting the radius's own slope as if it were the tangent line's slope.



Shortcut

Since the center (2,1) and point (5,5) differ by (3, 4) — a recognizable 3-4-5 relationship — you can read the radius's slope (4/3, i.e., 'rise 4, run 3') directly from the coordinate differences without formally writing the slope formula.



Desmos Tip

Type $y - 5 = (-3/4)(x - 5)$ into Desmos alongside the circle equation to visually confirm the line touches the circle at exactly one point, (5, 5).



Process of Elimination

Correct answer: **-3/4**

Tempting wrong answer	Exactly what went wrong
4/3	Reported the radius's slope — forgot the tangent is perpendicular.
3/4	Took the reciprocal but forgot to change the sign.
-4/3	Changed the sign but forgot to take the reciprocal.

Problem 7

Very Hard

Points A, B, and C lie on a circle, and AC is a diameter of the circle. If the measure of angle CAB is 35° , what is the measure of angle ACB?

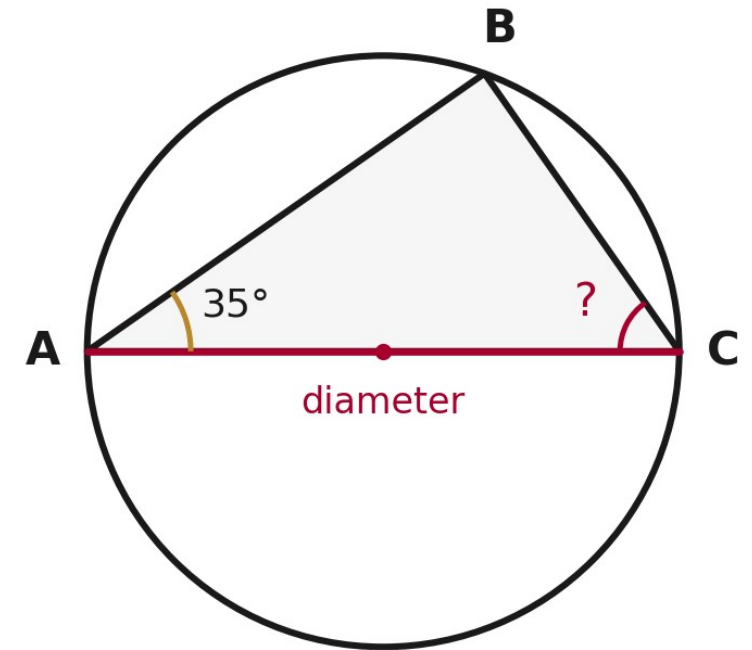
Formula / Rule

Thales' Theorem: an angle inscribed in a semicircle (with the diameter as one side of the triangle) is always 90° .

Solution

1. Since AC is a diameter and B lies on the circle, triangle ABC is inscribed in a semicircle with AC as the hypotenuse.
2. By Thales' Theorem, the angle opposite the diameter — angle ABC — must be exactly 90° .
3. Apply the Triangle Angle Sum Theorem: $\text{angle CAB} + \text{angle ABC} + \text{angle ACB} = 180$.
4. Substitute the known angles: $35 + 90 + \text{angle ACB} = 180$.
5. Combine: $125 + \text{angle ACB} = 180$.
6. Subtract 125 from both sides: $\text{angle ACB} = 55$.

✓ **Answer:** 55°





Problem 7 — Strategy & Elimination



Tip

The instant you see a triangle inscribed in a circle with one side explicitly labeled as a diameter, mark the angle opposite that diameter as 90° — this fact (Thales' Theorem) is almost never stated outright in the problem, but it's essential to solving it.



Trap

A common error is not recognizing that angle ABC (opposite the diameter AC) is 90° , and instead trying to solve the triangle with only one given angle and no way to find the third.



Shortcut

Since angle B is always 90° whenever AC is a diameter, the other two angles (at A and C) are automatically complementary to each other — so angle ACB = $90^\circ - \text{angle CAB} = 90^\circ - 35^\circ = 55^\circ$, without needing to write out the full 180° triangle sum.



Desmos Tip

Not a graphing question, but you can confirm the arithmetic by typing $90 - 35$ into Desmos to get 55 directly.



Process of Elimination

Correct answer: **55°**

Tempting wrong answer	Exactly what went wrong
90°	Reported angle ABC — the right angle.
35°	Assumed the two acute angles are equal.
145°	Computed $180^\circ - 35^\circ$ and ignored the 90° at B.

Problem 8

Very Hard

A sector of a circle with radius r has an area of 27π and a central angle of 60° . What is the value of r ?



Formula / Rule

Sector area = $(\theta/360)(\pi r^2)$.

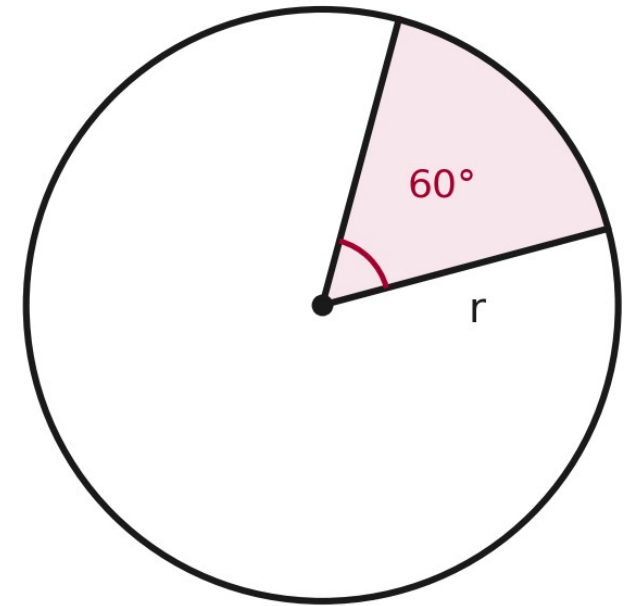


Solution

1. Substitute the given values into the sector area formula: $27\pi = (60/360)(\pi r^2)$.
2. Simplify the fraction $60/360 = 1/6$.
3. Rewrite the equation: $27\pi = (1/6)\pi r^2$.
4. Divide both sides by π : $27 = (1/6)r^2$.
5. Multiply both sides by 6: $r^2 = 162$.
6. Take the square root: $r = \sqrt{162}$.
7. Simplify the radical: $162 = 81 \times 2$, so $\sqrt{162} = \sqrt{81 \times 2} = 9\sqrt{2}$.



Answer: $9\sqrt{2}$



sector area = 27π



Problem 8 — Strategy & Elimination



Tip

When solving for the radius from a sector area, isolate r^2 completely first (clearing both the angle fraction and π) before taking the square root — trying to take a square root too early, while π or a fraction is still mixed in, invites errors.



Shortcut

Recognize that dividing out π immediately (since it appears on both sides) simplifies the equation right away to a plain algebra problem — $27 = (1/6)r^2$, which clears in two clean steps ($\times 6$, then $\sqrt{}$) without ever having to handle π alongside a square root.



Trap

A common error is taking the square root before fully isolating r^2 , for example incorrectly 'square-rooting through' the equation while π or the $1/6$ fraction is still attached, which produces a wrong expression entirely.



Desmos Tip

Type `sqrt(162)` into Desmos to confirm it equals $9\sqrt{2}$ (≈ 12.73).



Process of Elimination

Correct answer: $9\sqrt{2}$

Tempting wrong answer	Exactly what went wrong
162	Forgot the square root.
$3\sqrt{3}$	Ignored the $1/6$: $r^2 = 27$.
$3\sqrt{2}/2$ (≈ 2.12)	Divided by 6 instead of multiplying: $r^2 = 4.5$.

The 4 Skills at a Glance

Skill	First Move	Main Trap
Area and Volume	Break composite figures into known pieces. Similar solids: cube the scale factor for volume.	Assuming volume scales like length (k) instead of k^3 .
Lines, Angles, and Triangles	Name the relationship first: equal or supplementary? Mark isosceles / similarity clues immediately.	Treating similar as congruent, or pairing the wrong sides.
Right Triangles and Trigonometry	Label opposite / adjacent / hypotenuse for THE angle asked. Use $\sin \theta = \cos(90^\circ - \theta)$.	Extending the sin/cos shortcut to tangent.
Circles	Complete the square to reach standard form. Tangent slope = negative reciprocal of radius slope.	Reporting r^2 instead of r .

Golden Rules for Test Day

1

Mark the figure first

Isosceles triangle, parallel line, or diameter? Mark equal sides, equal angles, or the 90° angle before any algebra.

2

Similar = proportional

Area scales by k^2 , volume by k^3 — never by k itself.

3

Hypotenuse before SOH-CAH-TOA

Missing? Find it with the Pythagorean theorem first.

4

Inscribed = $\frac{1}{2}$ central

Same arc $\rightarrow$ the inscribed angle is exactly half. Never set them equal.

5

Desmos confirms, theory decides

Check answers numerically — but know which theorem applies.



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